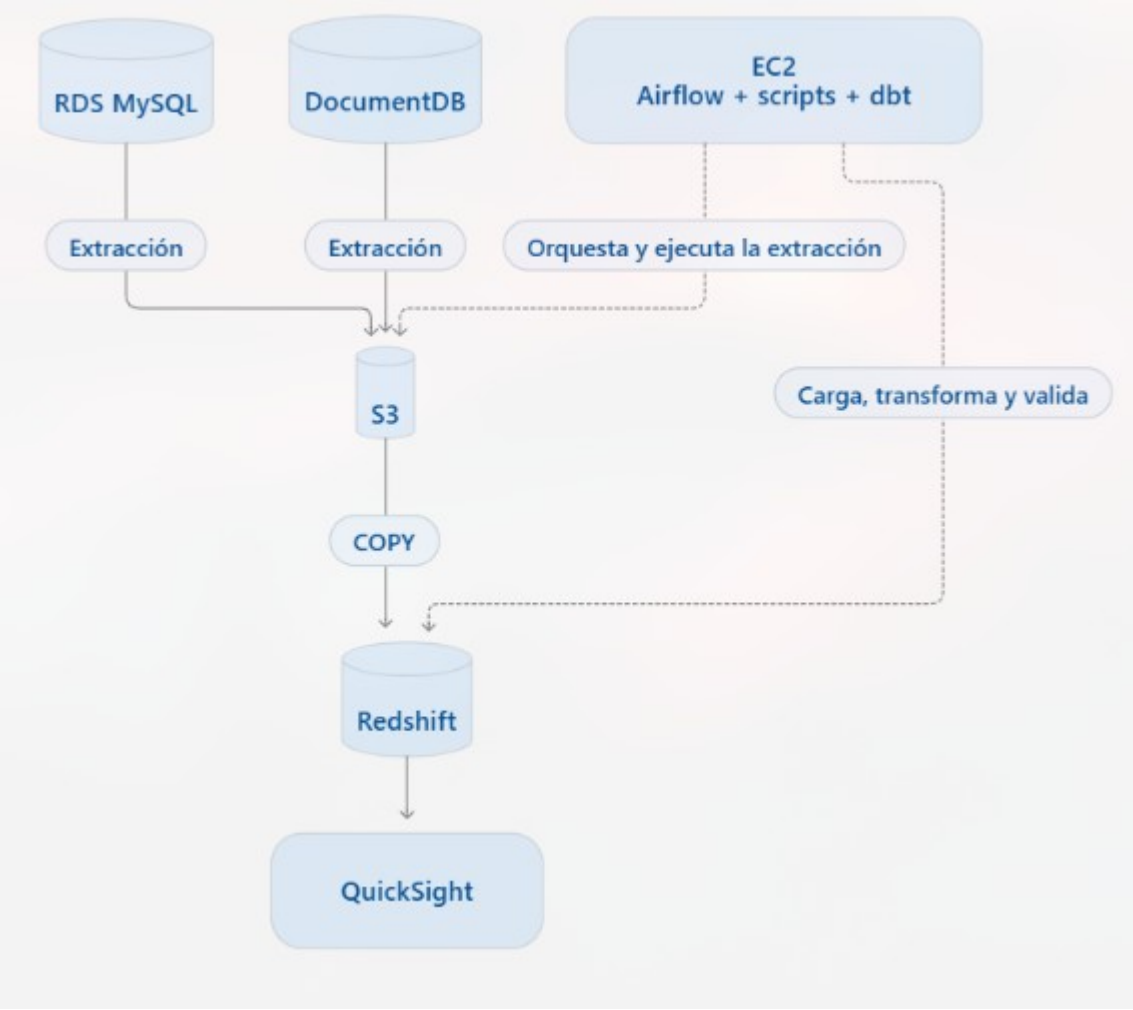
Taller Data Warehouse DE 2026-2

Presentado por: Johan Daniel Duque Hernández

Repositorio de GitHub: https://github.com/Dduquejh/Taller-DataWarehouse-DE-20262

En este trabajo se migra a AWS el repositorio https://github.com/smm-0216/DE-DW/tree/main que antes se ejecutaba en local, usando terraform para crear la infraestructura como código (IaC).

Arquitectura



Los datos salen de **Amazon RDS for MySQL** y **Amazon DocumentDB** y se guardan como archivos Parquet en un bucket privado de **Amazon S3**. Desde allí se cargan con COPY a **Amazon Redshift Serverless**, donde dbt construye las capas raw, intermediate, quarantine y marts. **Amazon QuickSight** consume una vista de marts. Apache Airflow y dbt corren en la misma instancia **Amazon EC2** con Docker Compose, y el SQL que envía dbt se ejecuta en Redshift.

Componente	Servicio y configuración	Función
Fuente relacional	RDS MySQL 8.4, db.t4g.micro	Clientes, productos, pedidos y líneas de pedido
Fuente documental	DocumentDB 5.0, db.t4g.medium	Reseñas y eventos de navegación
Almacenamiento	S3 privado con versionado	Conserva los lotes para poder reprocesarlos sin consultar las fuentes
Data warehouse	Redshift Serverless, 8 RPU	Capas del warehouse, modelo dimensional y SCD tipo 2
Orquestación	EC2 t3.medium con Airflow y dbt	Ejecuta y controla el pipeline
Consumo	QuickSight	Dashboard de ventas

Levantar la infraestructura

Se usa terraform para levantar todos los servicios con los permisos necesarios de IAM.

Todo el ambiente se define en un solo stack de Terraform, en la carpeta terraform/: un apply lo crea y un destroy lo elimina. La configuración va en terraform.tfvars, que no se sube al repositorio y se crea a partir de terraform.tfvars.example, un archivo de ejemplo sin secretos. Allí se definen la región (us-east-2), el perfil de AWS, el repositorio y la IP autorizada para entrar a Airflow.

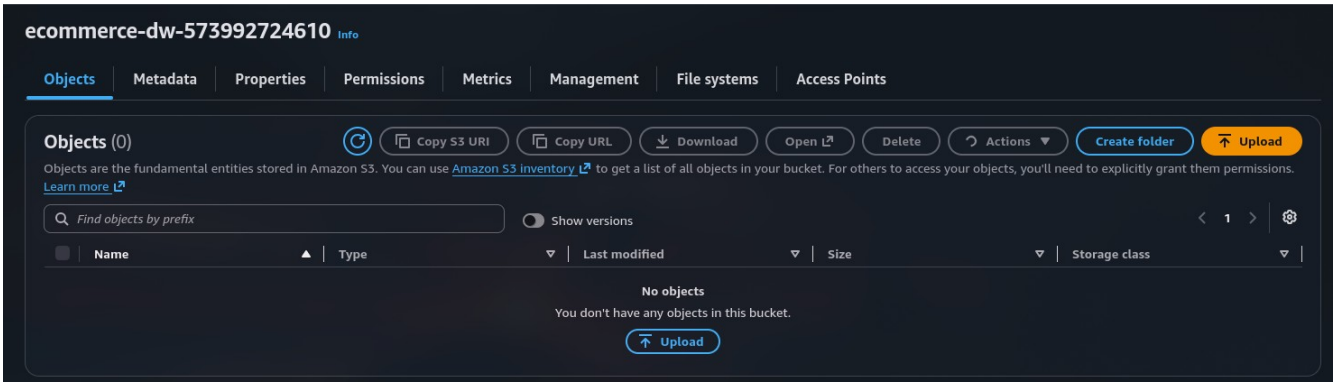
El plan crea 53 recursos. Son tantos porque Terraform cuenta cada pieza por separado, como cada regla de firewall o cada configuración del bucket; la mayoría no genera costo.

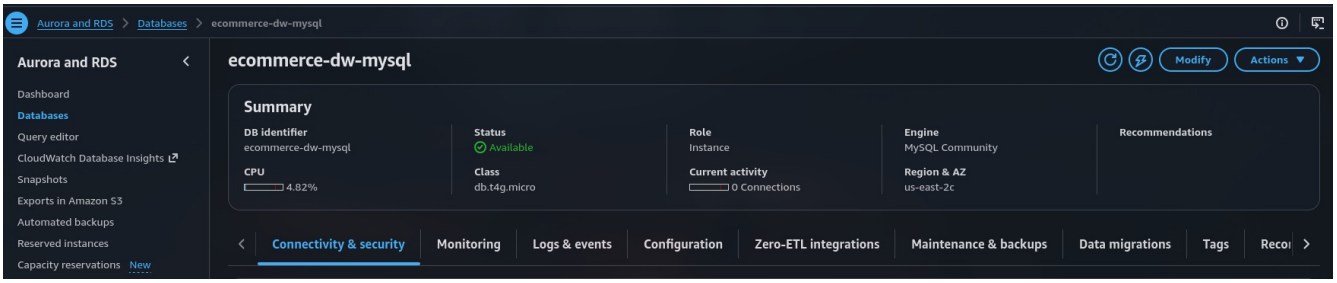
Grupo	Recursos	Qué incluye
Red	28	VPC, subred pública, 3 subredes privadas, rutas, endpoint de S3, 5 security groups y sus reglas
Fuentes	7	RDS MySQL y DocumentDB, con sus grupos de subredes y parámetros
S3	7	Bucket, bloqueo de acceso público, cifrado, versionado, ciclo de vida y política de TLS
IAM	6	Rol de la EC2 y rol de Redshift para leer S3
Redshift	3	Namespace, workgroup y límite diario de uso
EC2	2	Instancia y parámetro de SSM con la configuración del pipeline
Total	**53**	La conexión de QuickSight se agrega después con un segundo apply (3 recursos más)

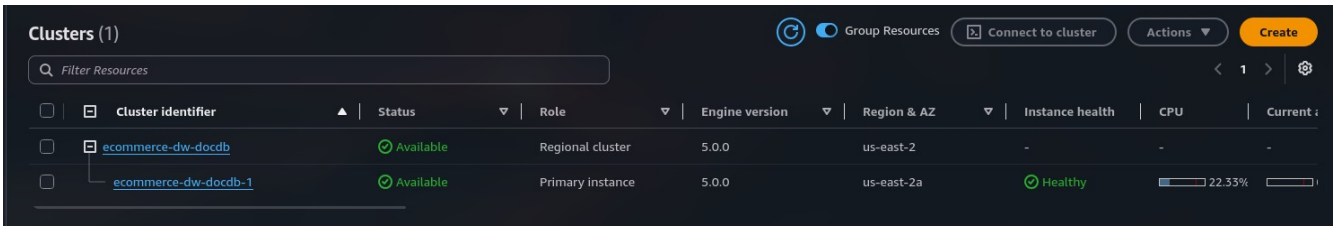
```
Apply complete! Resources: 53 added, 0 changed, 0 destroyed.

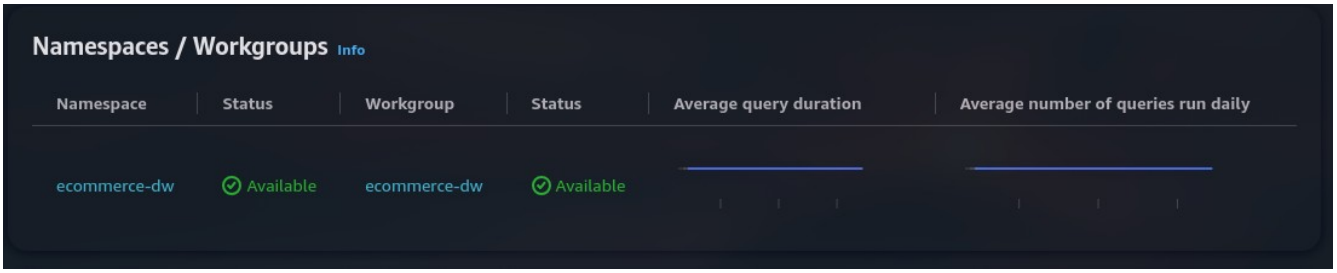
Outputs:

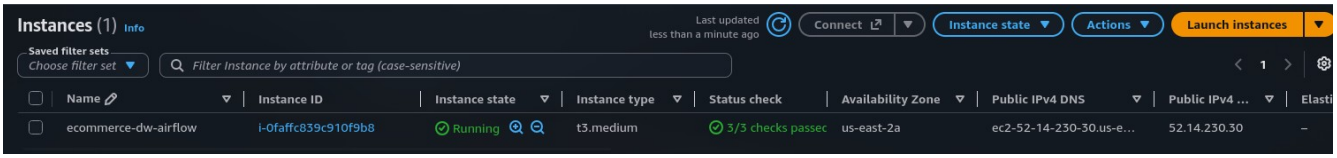
airflow_password_command = "sudo grep AIRFLOW_ADMIN_PASSWORD /opt/ecommerce-dw/.airflo
w-secrets"
airflow_url = "http://52.14.230.30:8080"
bucket_name = "ecommerce-dw-573992724610"
docdb_endpoint = "ecommerce-dw-docdb.cluster-c1aymgwc63hv.us-east-2.docdb.amazonaws.co
m"
instance_id = "i-0faffc839c910f9b8"
mysql_endpoint = "ecommerce-dw-mysql.c1aymgwc63hv.us-east-2.rds.amazonaws.com"
pipeline_config_parameter = "/ecommerce-dw/pipeline/env"
public_ip = "52.14.230.30"
redshift_copy_role_arn = "arn:aws:iam::573992724610:role/ecommerce-dw-redshift-s3"
redshift_endpoint = "ecommerce-dw.573992724610.us-east-2.redshift-serverless.amazonaws
.com"
secret_arns = {
  "docdb" = "arn:aws:secretsmanager:us-east-2:573992724610:secret:rds!cluster-3dda92fb
-051b-4e40-b4ae-eba7dcdfe0d1-DZdkdT"
  "mysql" = "arn:aws:secretsmanager:us-east-2:573992724610:secret:rds!db-9b0180c8-b0e6
-40e1-a1d6-9c8dc61957a6-laiRnW"
  "redshift" = "arn:aws:secretsmanager:us-east-2:573992724610:secret:redshift!ecommerc
e-dw-dw_admin-8rtk00"
}
secret_value_commands = {
  "docdb" = "aws secretsmanager get-secret-value --secret-id 'arn:aws:secretsmanager:u
s-east-2:573992724610:secret:rds!cluster-3dda92fb-051b-4e40-b4ae-eba7dcdfe0d1-DZdkdT'
--region us-east-2 --profile admin --query SecretString --output text"
  "mysql" = "aws secretsmanager get-secret-value --secret-id 'arn:aws:secretsmanager:u
s-east-2:573992724610:secret:rds!db-9b0180c8-b0e6-40e1-a1d6-9c8dc61957a6-laiRnW' --reg
ion us-east-2 --profile admin --query SecretString --output text"
  "redshift" = "aws secretsmanager get-secret-value --secret-id 'arn:aws:secretsmanage
r:us-east-2:573992724610:secret:redshift!ecommerce-dw-dw_admin-8rtk00' --region us-eas
t-2 --profile admin --query SecretString --output text"
```







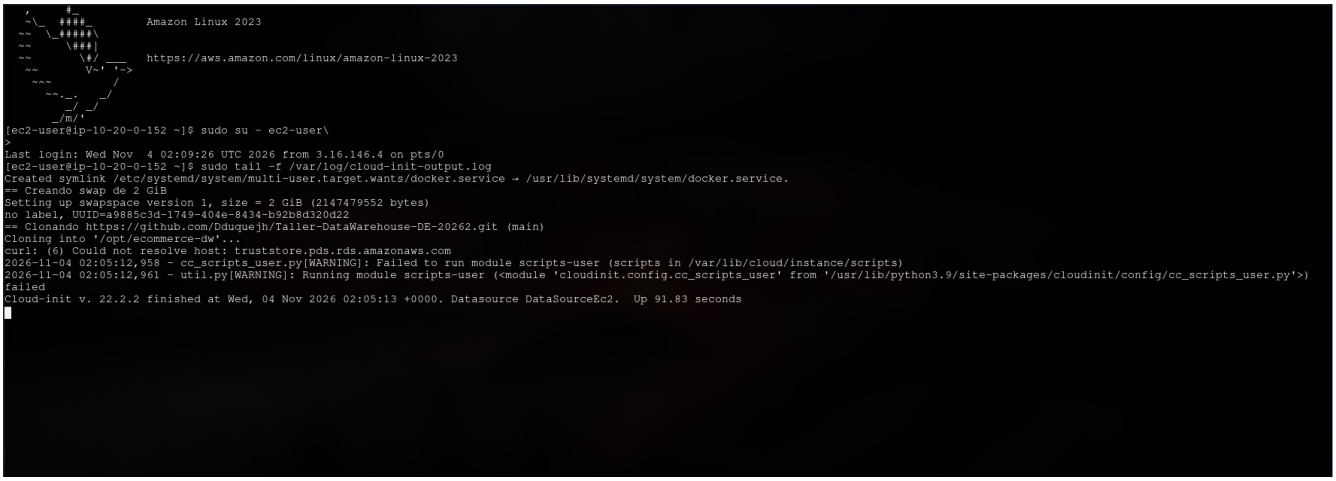




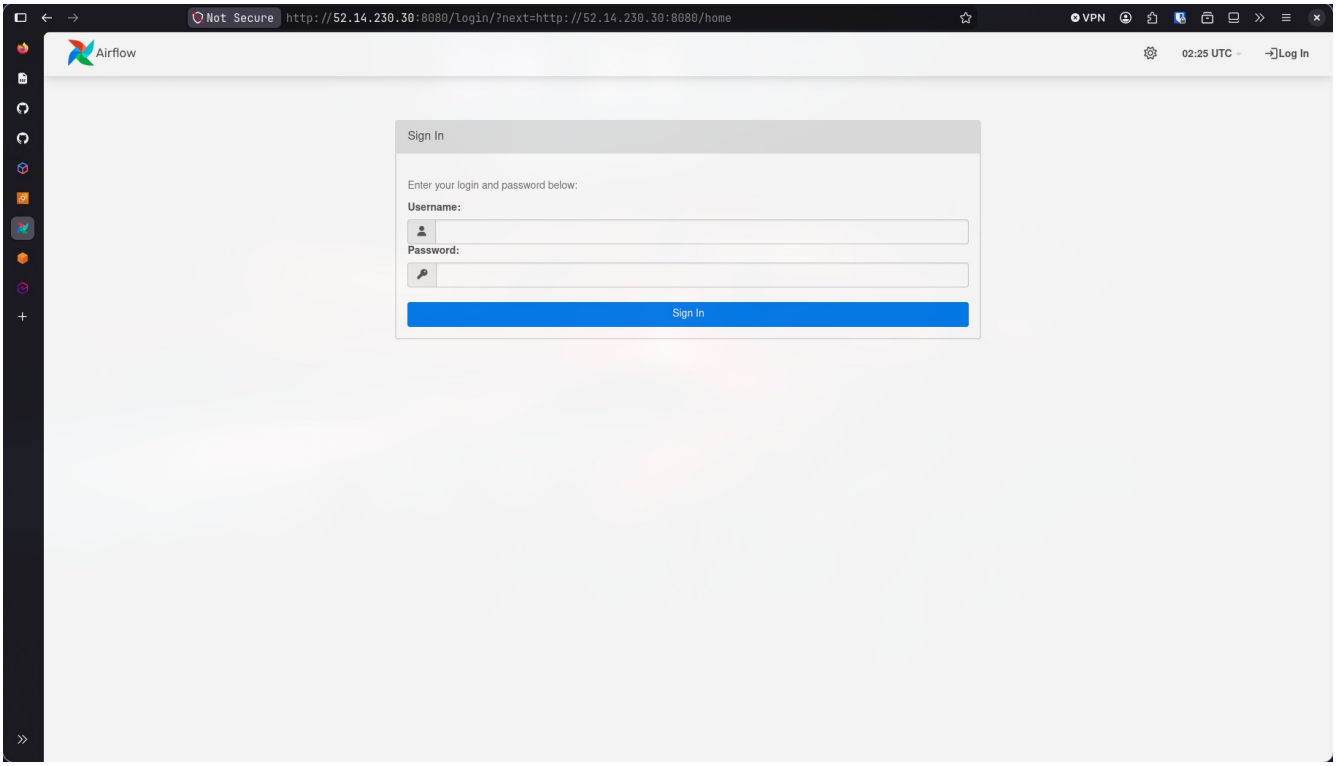
Arranque de la instancia y acceso

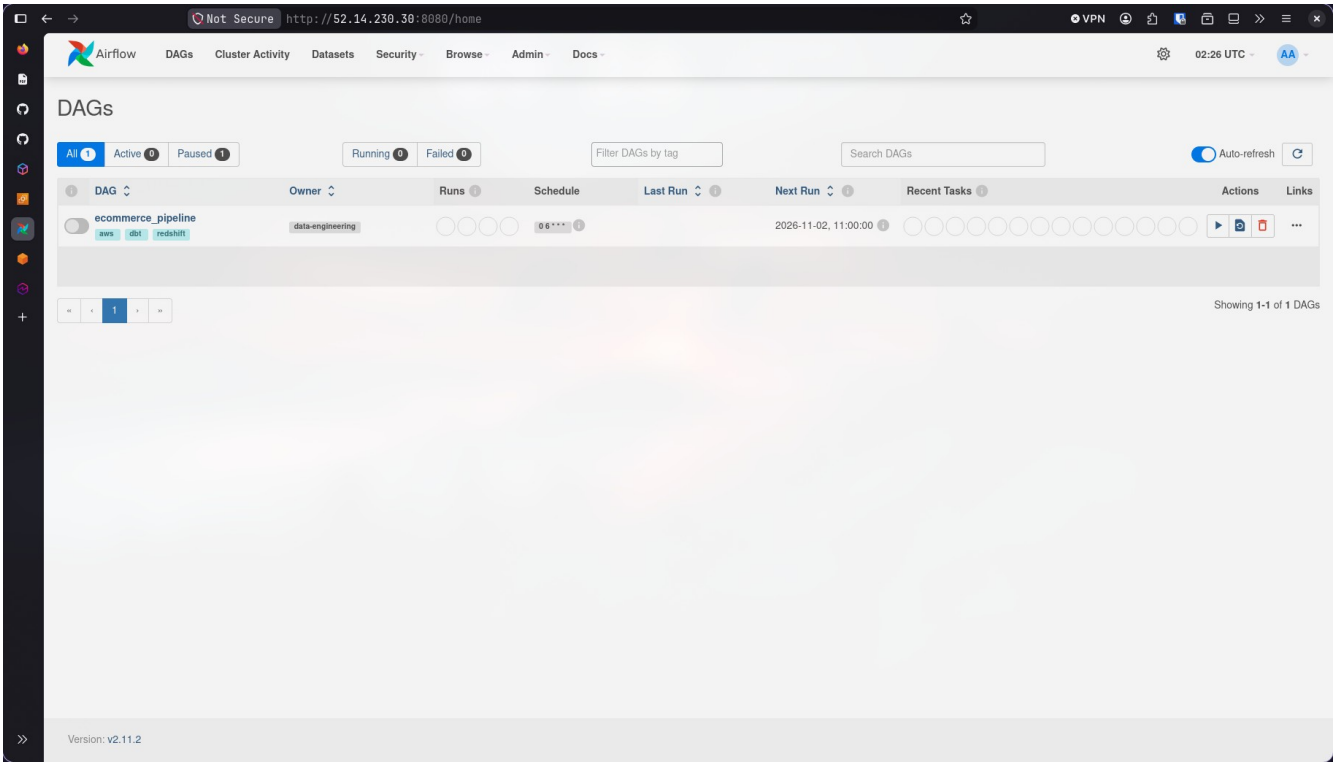
Al crearse, la instancia ejecuta un script de arranque que instala Docker, clona este repositorio, descarga el certificado TLS de AWS, genera el archivo .env desde SSM y levanta Airflow. Este proceso tarda entre 5 y 10 minutos después del apply.

La interfaz de Airflow se abre en la URL que entrega terraform output -raw airflow_url (puerto 8080), con el usuario airflow. El puerto solo acepta conexiones de la IP definida en allowed_cidrs, y la contraseña se genera en la instancia y se guarda en .airflow-secrets.



Se espera que termine de hacer la carga inicial del docker con airflow para poder acceder a la interfaz gráfica





Permisos y conexiones

Las bases de datos y Redshift no son públicas. Cada conexión está autorizada de forma explícita:

Origen	Destino	Puerto	Autorización
IP autorizada	Airflow (EC2)	8080	Regla de entrada limitada a allowed_cidrs
EC2 Instance Connect	EC2	22	Lista de prefijos administrada de ese servicio de AWS
Airflow (EC2)	RDS MySQL	3306	Security group de Airflow, Secrets Manager y TLS obligatorio
Airflow (EC2)	DocumentDB	27017	Security group de Airflow, Secrets Manager y TLS obligatorio
Airflow y dbt	Redshift	5439	Security group de Airflow, Secrets Manager y require_ssl
Airflow (EC2)	S3 raw/ y control/	443	Rol IAM de la instancia, sin DeleteObject en raw/
Redshift	S3 raw/	443	Rol IAM de solo lectura, por el endpoint gateway de S3
QuickSight	Redshift	5439	Conexión VPC con su propio security group

Decisiones técnicas

Decisión	Justificación
Parquet como formato en S3	COPY … FORMAT AS PARQUET conserva decimales, fechas y booleanos sin depender de separadores ni comillas, algo importante porque las reseñas traen texto libre y JSON.
Redshift Serverless	No requiere administrar un clúster y cobra solo mientras ejecuta consultas. Se dejó en 8 RPU con un límite de 16 RPU-hora al día para controlar el costo.
Un solo DAG	Reúne validación, extracción, carga, transformación, pruebas y registro del resultado, con dependencias explícitas y sin pasos manuales entre etapas.
raw de solo anexado	Cada lote se agrega con su _batch_id y staging toma la versión más reciente de cada llave. Así se conserva la trazabilidad y repetir un lote no duplica datos.
Lotes inmutables en S3	Las escrituras usan If-None-Match: * y el rol de la instancia no puede borrar en raw/. Un reintento reutiliza los archivos existentes y no vuelve a consultar la fuente.
Marca de agua al confirmar el COPY	Si dbt falla, los datos ya están seguros en raw y se puede volver a ejecutar sin extraer de nuevo y sin perder filas.
Un solo stack de Terraform	Permite levantar y eliminar todo el ambiente con un comando, lo que reduce el costo porque solo está encendido durante el trabajo.

Diferencias frente a la versión local

Aspecto	Local	AWS
Fuentes	MySQL 8.4 y MongoDB 8 en Docker	RDS MySQL 8.4 y DocumentDB 5.0, privados y con TLS obligatorio
Credenciales	Archivo .env	Secrets Manager y rol IAM de la instancia
Extracción	CSV por corrida, siempre completa	Parquet por lote en S3, completa o incremental
Capa raw	Se reemplazaba en cada corrida	Solo anexado, con _batch_id
Warehouse	DuckDB, un solo escritor	Redshift Serverless, dbt con 4 hilos
Carga	INSERT … FROM read_csv	COPY … FORMAT AS PARQUET con rol IAM
DAGs	Maestro, extracción y transformación	Un solo DAG con validación, reprocesamiento y registro del resultado
Snapshots	Esquema intermediate	Esquema snapshots

Ajustes de SQL para Redshift

DuckDB	Redshift
select * exclude (col)	dbt_utils.star(except=[...])
timezone('America/Bogota', ts)	Macro to_business_time, con convert_timezone
a is distinct from b	Macro is_distinct_from
extract(isodow from d)	Macro iso_day_of_week, a partir de dow

DuckDB	Redshift
json_extract_string(doc, '$.a.b')	json_extract_path_text(doc, 'a', 'b', true)
datediff('day', a, b), fecha - n	dbt.datediff, dbt.dateadd
TIMESTAMPTZ en raw	TIMESTAMP en UTC; la conversión se hace en los hechos
Esquema raw	RAW es palabra reservada, así que va entre comillas

Diferencias entre MongoDB y DocumentDB

- **Escrituras reintentables:** DocumentDB no las admite, así que la conexión usa retryWrites=False.
- **Conexión:** requiere TLS con global-bundle.pem, replicaSet=rs0 y authSource=admin.
- **Validador $jsonSchema:** DocumentDB 5.0 no admite todas las palabras clave. Si rechaza el validador, la colección se crea sin él, se registra una advertencia y la calidad se sigue controlando en dbt.
- **bypassDocumentValidation:** si no está disponible, los registros defectuosos se insertan sin esa opción.

Funcionamiento del pipeline

El DAG ecommerce_pipeline se puede ejecutar desde Airflow sin intervención manual entre etapas. Está programado a diario a las 06:00 (hora de Bogotá) y también se puede disparar a mano.

Etapa	Tarea	Qué hace
1	validate_and_prepare	Valida parámetros y conexiones, crea esquemas y tablas de control y calcula el lote
2-3	extract.mysql, extract.documentdb	Extraen en paralelo y escriben Parquet y un manifiesto en S3
4	load_raw	Ejecuta COPY del lote en una sola transacción y verifica los conteos contra el manifiesto
5	dbt_deps a dbt_run_marts	Ejecutan seeds, staging y calidad, cuarentena, snapshots SCD2 y marts
6	dbt_test	Ejecuta las pruebas de calidad y conciliación
7	record_result	Siempre se ejecuta: registra el resultado y marca el lote como processed si todo salió bien

Las tareas tienen por defecto 2 reintentos con espera exponencial de 30 segundos a 5 minutos. Las excepciones son validate_and_prepare, con 1 reintento, y dbt_test y record_result, sin reintentos porque repetirlas no cambiaría el resultado. max_active_runs=1 evita lotes concurrentes y cada línea de log lleva [batch=<id>], lo que permite identificar el lote y la tarea que fallaron.

Identificación de los lotes

- **batch_id:** es el instante lógico de la ejecución en UTC, por ejemplo 20261103T110000Z. Se mantiene igual en reintentos y al limpiar tareas.
- **En S3:** raw/<fuente>/<entidad>/batch_id=<id>/ contiene data.parquet y _manifest.json. El manifiesto registra el lote, el modo, la fecha de extracción, las marcas de agua, la cantidad de filas y el SHA-256.
- **En Redshift:** cada fila de raw lleva _batch_id, _source y _extracted_at, y control.batch_entities guarda una fila por lote y entidad.

Estados y marca de avance

Estado	Dónde se registra	Cuándo cambia
Extraído	Manifiesto en S3	Al escribir el Parquet y el manifiesto
loaded	control.batch_entities	En el mismo COMMIT del COPY. **Desde aquí avanza la marca de agua**
processed	control.batch_entities	Cuando dbt (modelos, snapshots y pruebas) terminó bien
success o failed	control.pipeline_runs	Al final de la ejecución, junto con las tareas fallidas

La extracción incremental usa una marca de agua por entidad:

Entidad	Columna de avance	Condición
customers, products	updated_at	updated_at > marca
orders, order_items	order_id, order_item_id	id > marca (inmutables)
reviews, navigation_events	review_id, event_id	id > marca (inmutables)

Evidencias de funcionamiento

En esta sección se presentan los resultados de la ejecución en AWS. Los comandos para reproducir cada escenario están en el README del repositorio.

Carga inicial

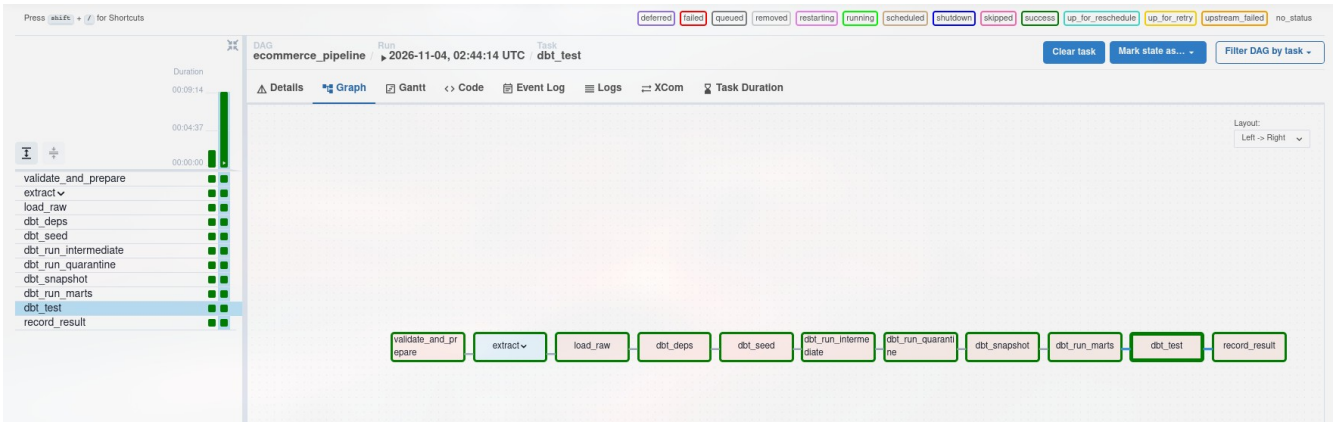
Para la carga inicial, primero se crean las tablas de RDS y las colecciones de DocumentDB, y se cargan los datos sintéticos, incluidos algunos registros defectuosos a propósito. Después se ejecuta el DAG en modo de carga completa, que extrae todas las entidades a S3, las carga en Redshift con COPY, ejecuta las transformaciones de dbt y corre las pruebas de calidad, sin intervención manual entre etapas. Se espera que los datos queden disponibles en los modelos analíticos y que los hechos concilien con la fuente validada.

Se generan los datos en las fuentes

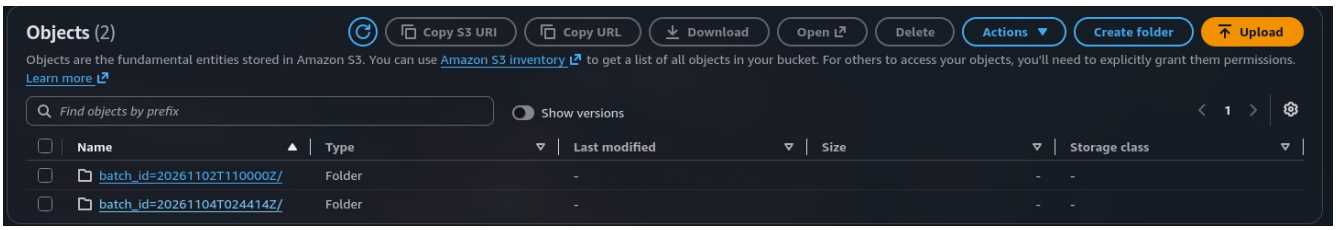
```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/generate_data.sh
Generando datos en RDS MySQL y DocumentDB...
INFO Found credentials from IAM Role: ecommerce-dw-airflow-ec2
INFO You appear to be connected to a DocumentDB cluster. For more information regarding feature compatibility and support please visit https://www.mongodb.com/supportability/documentdb
MySQL: 50 customers, 25 products, 100 orders, 253 order items
MongoDB: 150 reviews, 500 navigation events; seed=42
Injected bad rows: 34 (--bad-records=2); includes valid order headers that isolate subtotal defects
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

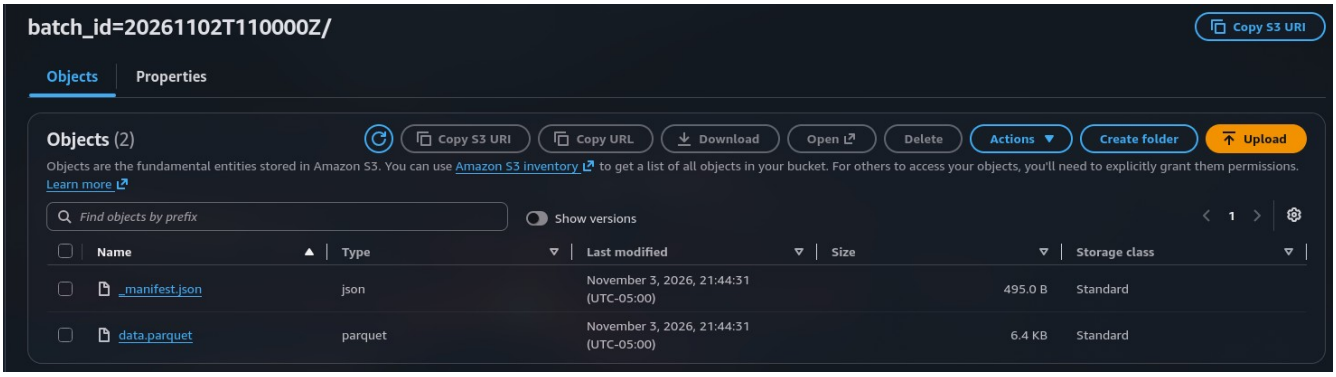
El DAG termina correctamente todas sus estapas

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/run_pipeline.sh --mode full
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a supported version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a supported version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
DAG ecommerce_pipeline disparado: run_id=manual__20261104T024358 conf={"mode": "full", "reprocess_batch_id": ""}
jq: parse error: Invalid numeric literal at line 1, column 15
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```










```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh runs
batch_id          dag_run_id                          mode  reprocess  status   failed_tasks  raw_rows  dbt_tests_passed  dbt_tests_failed  finished_at
----------------  ----------------------------------  ----  ---------  -------  ------------  --------  ----------------  ----------------  -------------------
20261104T024414Z  manual__20261104T024358              full  False      success  None          1112      107               None              2026-11-04 03:01:33
20261102T110000Z  scheduled__2026-11-02T11:00:00+00:00  auto  False      success  None          1112      107               None              2026-11-04 02:46:15
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh batches
batch_id          raw_table                mode  watermark_from  watermark_to         row_count  status     loaded_at            processed_at
----------------  -----------------------  ----  --------------  -------------------  ---------  ---------  -------------------  -------------------
20261104T024414Z  mongo_navigation_events  full  None            800007               508        processed  2026-11-04 02:52:37  2026-11-04 03:01:31
20261104T024414Z  mongo_reviews            full  None            800007               158        processed  2026-11-04 02:52:38  2026-11-04 03:01:31
20261104T024414Z  mysql_customers          full  None            2025-01-01 00:00:00  54         processed  2026-11-04 02:52:39  2026-11-04 03:01:31
20261104T024414Z  mysql_order_items        full  None            800005               259        processed  2026-11-04 02:52:40  2026-11-04 03:01:31
20261104T024414Z  mysql_orders             full  None            800005               106        processed  2026-11-04 02:52:40  2026-11-04 03:01:31
20261104T024414Z  mysql_products           full  None            2025-01-01 00:00:00  27         processed  2026-11-04 02:52:41  2026-11-04 03:01:31
20261102T110000Z  mongo_navigation_events  full  None            800007               508        processed  2026-11-04 02:44:37  2026-11-04 02:46:14
20261102T110000Z  mongo_reviews            full  None            800007               158        processed  2026-11-04 02:44:39  2026-11-04 02:46:14
20261102T110000Z  mysql_customers          full  None            2025-01-01 00:00:00  54         processed  2026-11-04 02:44:40  2026-11-04 02:46:14
20261102T110000Z  mysql_order_items        full  None            800005               259        processed  2026-11-04 02:44:41  2026-11-04 02:46:14
20261102T110000Z  mysql_orders             full  None            800005               106        processed  2026-11-04 02:44:43  2026-11-04 02:46:14
20261102T110000Z  mysql_products           full  None            2025-01-01 00:00:00  27         processed  2026-11-04 02:44:44  2026-11-04 02:46:14
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh reconcile
fact                 rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
-------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales            253   253            7293790.26  42c56f2fe5ba90f5  253                7293790.26
fct_reviews          150   150            0           2af153c362df1946  150                0
fct_navigation_events  500   500            0           53182146b1f9240e  500                0
OK: cada hecho coincide con su fuente validada y no tiene llaves duplicadas
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

Registros defectuosos

Resultado esperado: rechazos identificados con su motivo y su lote de origen.

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh quarantine
entity                        rejection_reason       _batch_id         rows
----------------------------  ---------------------  ----------------  ----
documentdb.navigation_events  invalid_event_type     20261104T024414Z  2
documentdb.navigation_events  missing_occurred_at    20261104T024414Z  2
documentdb.navigation_events  orphan_customer        20261104T024414Z  2
documentdb.navigation_events  orphan_product         20261104T024414Z  2
documentdb.reviews            invalid_rating         20261104T024414Z  2
documentdb.reviews            missing_reviewed_at    20261104T024414Z  2
documentdb.reviews            orphan_customer        20261104T024414Z  2
documentdb.reviews            orphan_product         20261104T024414Z  2
mysql.customers               invalid_email          20261104T024414Z  2
mysql.customers               missing_name           20261104T024414Z  2
mysql.order_items             orphan_order           20261104T024414Z  2
mysql.order_items             orphan_product         20261104T024414Z  2
mysql.order_items             subtotal_mismatch      20261104T024414Z  2
mysql.orders                  invalid_sales_channel  20261104T024414Z  2
mysql.orders                  orphan_customer        20261104T024414Z  2
mysql.products                missing_category       20261104T024414Z  2
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

Cambio en un atributo historizado (SCD tipo 2)

Resultado esperado: una versión nueva con is_current = true y la anterior cerrada con valid_to.

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/simulate_changes.sh
Aplicando cambios en la fuente...
Customer 1: Bogotá, Bogotá, D.C. -> Medellín, Antioquia
Product 1: Electrónica -> Hogar
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/run_pipeline.sh
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a supported version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a supported version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
DAG ecommerce_pipeline disparado: run_id=manual__20261104T030724 conf={"mode": "auto", "reprocess_batch_id": ""}
jq: parse error: Invalid numeric literal at line 1, column 15
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh scd2
dim_customer, customer_id=1
customer_sk                       city    department    valid_from           valid_to  is_current
--------------------------------  ------  ------------  -------------------  --------  ----------
aff00fde234279a9064828b48dbe629f  Bogotá  Bogotá, D.C.  2024-11-10 00:00:00  None      True

dim_product, product_id=1
product_sk                        category     valid_from           valid_to  is_current
--------------------------------  -----------  -------------------  --------  ----------
4fe90e2ee856c44d4857ea4f72a5a811  Electrónica  2025-01-01 00:00:00  None      True
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```



Nuevo lote incremental

Resultado esperado: cada hecho crece en 3 filas, y repetir el lote o reprocesarlo no cambia nada.

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/verify_incremental_load.sh --batch-id 2
fact                 rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
-------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales            256   256            7357260.42  8896681228489ff3  256               7357260.42
fct_reviews          153   153            0           4fac51cf49505be9  153               0
fct_navigation_events  503   503            0           0cad4c3b99974cb9  503               0
OK: cada hecho coincide con su fuente validada y no tiene llaves duplicadas
Estado inicial:
fct_sales 256 256 7357260.42 8896681228489ff3
fct_reviews 153 153 0 4fac51cf49505be9
fct_navigation_events 503 503 0 0cad4c3b99974cb9

/opt/airflow/src/ecommerce_dw/connections.py:99: UserWarning: You appear to be connected to a DocumentDB cluster. For more information regarding feature compatibility and support please visit ht
tps://www.mongodb.com/supportability/documentdb
  return MongoClient(**options)
Agregando a las fuentes el lote diario 2...
Batch 2 (2026-01-02): 3 new orders, 3 reviews and 3 navigation events upserted
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
DAG ecommerce_pipeline disparado: run_id=manual__20261104T031526 conf={"mode": "auto", "reprocess_batch_id": ""}
Resultado de manual__20261104T031526: success
batch_id             dag_run_id                        mode  reprocess  status   failed_tasks  raw_rows  dbt_tests_passed  dbt_tests_failed  finished_at
-------------------  --------------------------------  ----  ---------  -------  ------------  --------  ----------------  ----------------  -------------------
20261104T031534Z  manual__20261104T031526            auto  False      success  None          12        107               None              2026-11-04 03:17:37
20261104T031040Z  manual__20261104T031030            auto  False      success  None          12        107               None              2026-11-04 03:12:39
fact                 rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
-------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales            259   259            7420730.58  bd515a5603aa1419  259               7420730.58
fct_reviews          156   156            0           8b57ea5592874612  156               0
fct_navigation_events  506   506            0           ce67e63a11c81b5a  506               0
OK: cada hecho coincide con su fuente validada y no tiene llaves duplicadas
OK lote nuevo: cada hecho creció en 3 filas
fct_sales 259 259 7420730.58 bd515a5603aa1419
fct_reviews 156 156 0 8b57ea5592874612
fct_navigation_events 506 506 0 ce67e63a11c81b5a

/opt/airflow/src/ecommerce_dw/connections.py:99: UserWarning: You appear to be connected to a DocumentDB cluster. For more information regarding feature compatibility and support please visit ht
tps://www.mongodb.com/supportability/documentdb
  return MongoClient(**options)
Agregando a las fuentes el lote diario 2...
Batch 2 (2026-01-02): 0 new orders, 3 reviews and 3 navigation events upserted
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
```

```
DAG ecommerce_pipeline disparado: run_id=manual__20261104T031757 conf={"mode": "auto", "reprocess_batch_id": ""}
Resultado de manual__20261104T031757: success
batch_id          dag_run_id                              mode  reprocess  status   failed_tasks  raw_rows  dbt_tests_passed  dbt_tests_failed  finished_at
----------------  --------------------------------------  ----  ---------  -------  ------------  --------  ----------------  ----------------  --------------------
20261104T031806Z  manual__20261104T031757                  auto  False      success  None          0         107               None              2026-11-04 03:20:06
20261104T031534Z  manual__20261104T031526                  auto  False      success  None          12        107               None              2026-11-04 03:17:37
OK repetición del lote en las fuentes: hechos sin cambios

/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
DAG ecommerce_pipeline disparado: run_id=manual__20261104T032025 conf={"mode": "auto", "reprocess_batch_id": "20261104T031806Z"}
Resultado de manual__20261104T032025: success
batch_id          dag_run_id                              mode       reprocess  status   failed_tasks  raw_rows  dbt_tests_passed  dbt_tests_failed  finished_at
----------------  --------------------------------------  ---------  ---------  -------  ------------  --------  ----------------  ----------------  --------------------
20261104T031806Z  manual__20261104T032025                  reprocess  True       success  None          0         107               None              2026-11-04 03:22:21
20261104T031806Z  manual__20261104T031757                  auto       False      success  None          0         107               None              2026-11-04 03:20:06
OK reprocesamiento desde S3 del lote 20261104T031806Z: hechos sin cambios
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```



Repetición de un lote exitoso

Resultado esperado: la misma huella (digest) en los tres hechos antes y después, y sin versiones históricas nuevas.

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh batches
batch_id          raw_table                mode         watermark_from       watermark_to         row_count  status     loaded_at            processed_at
----------------  -----------------------  -----------  -------------------  -------------------  ---------  ---------  -------------------  -------------------
20261104T031806Z  mongo_navigation_events  incremental  1000022              1000022              0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mongo_reviews            incremental  1000022              1000022              0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mysql_customers          incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mysql_order_items        incremental  10000022             10000022             0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mysql_orders             incremental  1000022              1000022              0          processed  2026-11-04 03:20:43  2026-11-04 03:22:21
20261104T031806Z  mysql_products           incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:20:43  2026-11-04 03:22:21
20261104T031534Z  mongo_navigation_events  incremental  1000012              1000022              3          processed  2026-11-04 03:15:49  2026-11-04 03:17:37
20261104T031534Z  mongo_reviews            incremental  1000012              1000022              3          processed  2026-11-04 03:15:50  2026-11-04 03:17:37
20261104T031534Z  mysql_customers          incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:15:50  2026-11-04 03:17:37
20261104T031534Z  mysql_order_items        incremental  10000012             10000022             3          processed  2026-11-04 03:15:51  2026-11-04 03:17:37
20261104T031534Z  mysql_orders             incremental  1000012              1000022              3          processed  2026-11-04 03:15:51  2026-11-04 03:17:37
20261104T031534Z  mysql_products           incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:15:52  2026-11-04 03:17:37
20261104T031040Z  mongo_navigation_events  incremental  800007               1000012              3          processed  2026-11-04 03:10:53  2026-11-04 03:12:39
20261104T031040Z  mongo_reviews            incremental  800007               1000012              3          processed  2026-11-04 03:10:55  2026-11-04 03:12:39
20261104T031040Z  mysql_customers          incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:10:55  2026-11-04 03:12:39
20261104T031040Z  mysql_order_items        incremental  800005               10000012             3          processed  2026-11-04 03:10:56  2026-11-04 03:12:39
20261104T031040Z  mysql_orders             incremental  800005               1000012              3          processed  2026-11-04 03:10:56  2026-11-04 03:12:39
20261104T031040Z  mysql_products           incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:10:57  2026-11-04 03:12:39
20261104T030732Z  mongo_navigation_events  incremental  800007               800007               0          processed  2026-11-04 03:07:45  2026-11-04 03:09:30
20261104T030732Z  mongo_reviews            incremental  800007               800007               0          processed  2026-11-04 03:07:46  2026-11-04 03:09:30
20261104T030732Z  mysql_customers          incremental  2025-01-01 00:00:00  2025-07-01 00:00:00  1          processed  2026-11-04 03:07:47  2026-11-04 03:09:30
20261104T030732Z  mysql_order_items        incremental  800005               800005               0          processed  2026-11-04 03:07:47  2026-11-04 03:09:30
20261104T030732Z  mysql_orders             incremental  800005               800005               0          processed  2026-11-04 03:07:47  2026-11-04 03:09:30
20261104T030732Z  mysql_products           incremental  2025-01-01 00:00:00  2025-07-01 00:00:00  1          processed  2026-11-04 03:07:48  2026-11-04 03:09:30
20261104T024414Z  mongo_navigation_events  full         None                 800007               508        processed  2026-11-04 02:52:37  2026-11-04 03:01:31
20261104T024414Z  mongo_reviews            full         None                 800007               158        processed  2026-11-04 02:52:38  2026-11-04 03:01:31
20261104T024414Z  mysql_customers          full         None                 2025-01-01 00:00:00  54         processed  2026-11-04 02:52:39  2026-11-04 03:01:31
20261104T024414Z  mysql_order_items        full         None                 800005               259        processed  2026-11-04 02:52:40  2026-11-04 03:01:31
20261104T024414Z  mysql_orders             full         None                 800005               106        processed  2026-11-04 02:52:40  2026-11-04 03:01:31
20261104T024414Z  mysql_products           full         None                 2025-01-01 00:00:00  27         processed  2026-11-04 02:52:41  2026-11-04 03:01:31
20261102T110000Z  mongo_navigation_events  full         None                 800007               508        processed  2026-11-04 02:44:37  2026-11-04 02:46:14
20261102T110000Z  mongo_reviews            full         None                 800007               158        processed  2026-11-04 02:44:39  2026-11-04 02:46:14
20261102T110000Z  mysql_customers          full         None                 2025-01-01 00:00:00  54         processed  2026-11-04 02:44:40  2026-11-04 02:46:14
20261102T110000Z  mysql_order_items        full         None                 800005               259        processed  2026-11-04 02:44:41  2026-11-04 02:46:14
20261102T110000Z  mysql_orders             full         None                 800005               106        processed  2026-11-04 02:44:43  2026-11-04 02:46:14
20261102T110000Z  mysql_products           full         None                 2025-01-01 00:00:00  27         processed  2026-11-04 02:44:44  2026-11-04 02:46:14
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh state
fact                rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales           259   259            7420730.58  bd515a5603aa1419  259               7420730.58
fct_reviews         156   156            0           8b57ea5592874612  156               0
fct_navigation_events  506   506            0           ce67e63a11c81b5a  506               0
```

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh batches
batch_id          raw_table               mode         watermark_from       watermark_to         row_count  status     loaded_at            processed_at
----------------  ----------------------  -----------  -------------------  -------------------  ---------  ---------  -------------------  -------------------
20261104T031806Z  mongo_navigation_events  incremental  1000022              1000022              0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mongo_reviews            incremental  1000022              1000022              0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mysql_customers          incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mysql_order_items        incremental  10000022             10000022             0          processed  2026-11-04 03:20:42  2026-11-04 03:22:21
20261104T031806Z  mysql_orders             incremental  1000022              1000022              0          processed  2026-11-04 03:20:43  2026-11-04 03:22:21
20261104T031806Z  mysql_products           incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:20:43  2026-11-04 03:22:21
20261104T031534Z  mongo_navigation_events  incremental  1000012              1000022              3          processed  2026-11-04 03:15:49  2026-11-04 03:17:37
20261104T031534Z  mongo_reviews            incremental  1000012              1000022              3          processed  2026-11-04 03:15:50  2026-11-04 03:17:37
20261104T031534Z  mysql_customers          incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:15:50  2026-11-04 03:17:37
20261104T031534Z  mysql_order_items        incremental  10000012             10000022             3          processed  2026-11-04 03:15:51  2026-11-04 03:17:37
20261104T031534Z  mysql_orders             incremental  1000012              1000022              3          processed  2026-11-04 03:15:51  2026-11-04 03:17:37
20261104T031534Z  mysql_products           incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:15:52  2026-11-04 03:17:37
20261104T031040Z  mongo_navigation_events  incremental  800007               1000012              3          processed  2026-11-04 03:10:53  2026-11-04 03:12:39
20261104T031040Z  mongo_reviews            incremental  800007               1000012              3          processed  2026-11-04 03:10:55  2026-11-04 03:12:39
20261104T031040Z  mysql_customers          incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:10:55  2026-11-04 03:12:39
20261104T031040Z  mysql_order_items        incremental  800005               10000012             3          processed  2026-11-04 03:10:56  2026-11-04 03:12:39
20261104T031040Z  mysql_orders             incremental  800005               1000012              3          processed  2026-11-04 03:10:56  2026-11-04 03:12:39
20261104T031040Z  mysql_products           incremental  2025-07-01 00:00:00  2025-07-01 00:00:00  0          processed  2026-11-04 03:10:57  2026-11-04 03:12:39
20261104T030732Z  mongo_navigation_events  incremental  800007               800007               0          processed  2026-11-04 03:07:45  2026-11-04 03:09:30
20261104T030732Z  mongo_reviews            incremental  800007               800007               0          processed  2026-11-04 03:07:46  2026-11-04 03:09:30
20261104T030732Z  mysql_customers          incremental  2025-01-01 00:00:00  2025-07-01 00:00:00  1          processed  2026-11-04 03:07:47  2026-11-04 03:09:30
20261104T030732Z  mysql_order_items        incremental  800005               800005               0          processed  2026-11-04 03:07:47  2026-11-04 03:09:30
20261104T030732Z  mysql_orders             incremental  800005               800005               0          processed  2026-11-04 03:07:47  2026-11-04 03:09:30
20261104T030732Z  mysql_products           incremental  2025-01-01 00:00:00  2025-07-01 00:00:00  1          processed  2026-11-04 03:07:48  2026-11-04 03:09:30
20261104T024414Z  mongo_navigation_events  full         None                 800007               508        processed  2026-11-04 02:52:37  2026-11-04 03:01:31
20261104T024414Z  mongo_reviews            full         None                 800007               158        processed  2026-11-04 02:52:38  2026-11-04 03:01:31
20261104T024414Z  mysql_customers          full         None                 2025-01-01 00:00:00  54         processed  2026-11-04 02:52:39  2026-11-04 03:01:31
20261104T024414Z  mysql_order_items        full         None                 800005               259        processed  2026-11-04 02:52:40  2026-11-04 03:01:31
20261104T024414Z  mysql_orders             full         None                 800005               106        processed  2026-11-04 02:52:40  2026-11-04 03:01:31
20261104T024414Z  mysql_products           full         None                 2025-01-01 00:00:00  27         processed  2026-11-04 02:52:41  2026-11-04 03:01:31
20261102T110000Z  mongo_navigation_events  full         None                 800007               508        processed  2026-11-04 02:44:37  2026-11-04 02:46:14
20261102T110000Z  mongo_reviews            full         None                 800007               158        processed  2026-11-04 02:44:39  2026-11-04 02:46:14
20261102T110000Z  mysql_customers          full         None                 2025-01-01 00:00:00  54         processed  2026-11-04 02:44:40  2026-11-04 02:46:14
20261102T110000Z  mysql_order_items        full         None                 800005               259        processed  2026-11-04 02:44:41  2026-11-04 02:46:14
20261102T110000Z  mysql_orders             full         None                 800005               106        processed  2026-11-04 02:44:43  2026-11-04 02:46:14
20261102T110000Z  mysql_products           full         None                 2025-01-01 00:00:00  27         processed  2026-11-04 02:44:44  2026-11-04 02:46:14
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh state
fact                rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales           259   259            7420730.58  bd515a5603aa1419  259                7420730.58
fct_reviews         156   156            0           8b57ea5592874612  156                0
fct_navigation_events  506   506            0           ce67e63a11c81b5a  506                0
```

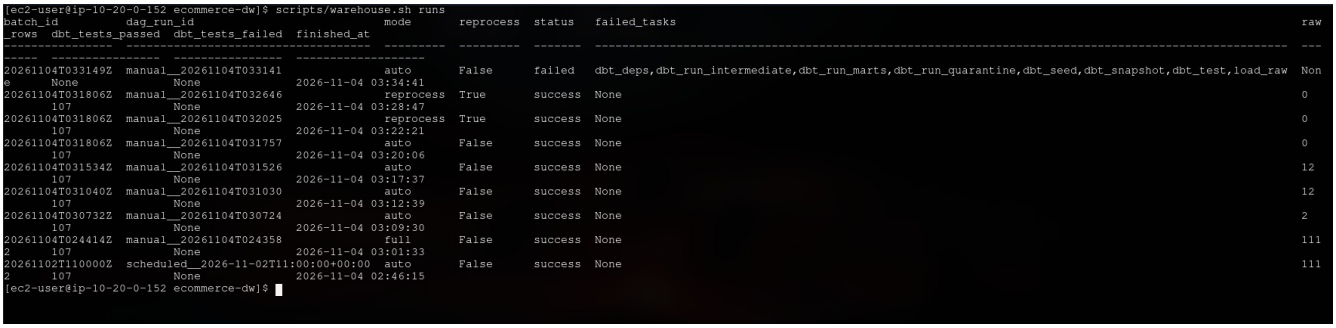
Fallo controlado y reejecución

Resultado esperado: recuperación sin pérdida ni duplicación. El fallo se provoca con una variable de Airflow y se identifica con el log de la tarea y con control.pipeline_runs:

- En **Admin → Variables** se crea ecommerce_dw_fail_stage con el valor load.
- Se ejecuta scripts/run_pipeline.sh. La tarea load_raw falla antes del COMMIT, así que la transacción se revierte y raw y control quedan como estaban.
- scripts/warehouse.sh runs muestra el batch_id, failed_tasks y el estado failed, y el log muestra [batch=…] fallo simulado.
- Se borra la variable y, en la ejecución fallida, se hace **Clear** sobre load_raw con Downstream activado. El lote conserva su batch_id, no se vuelve a extraer porque los archivos ya están en S3 y se carga una sola vez.
- scripts/warehouse.sh reconcile y state comprueban que no hubo pérdida ni duplicados.







```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh runs
batch_id          dag_run_id                                  mode       reprocess  status   failed_tasks                                                
```

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

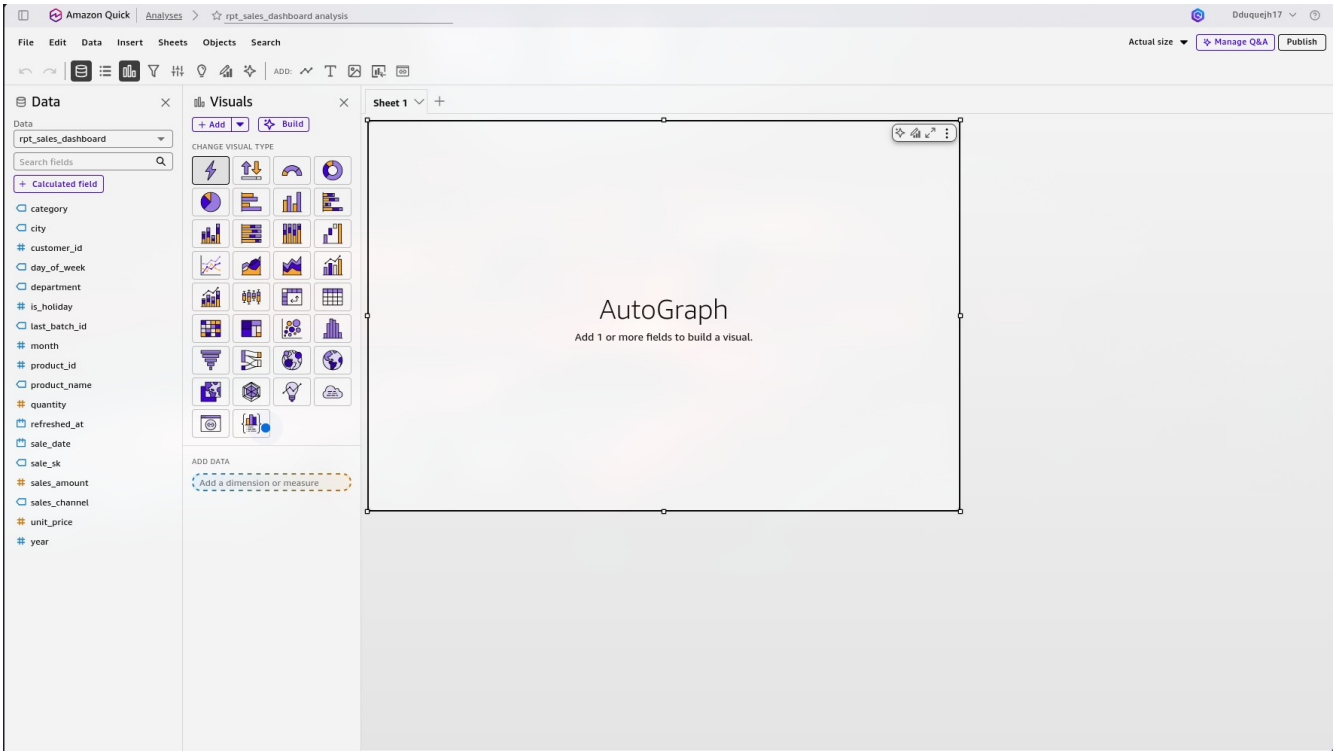
Outputs:

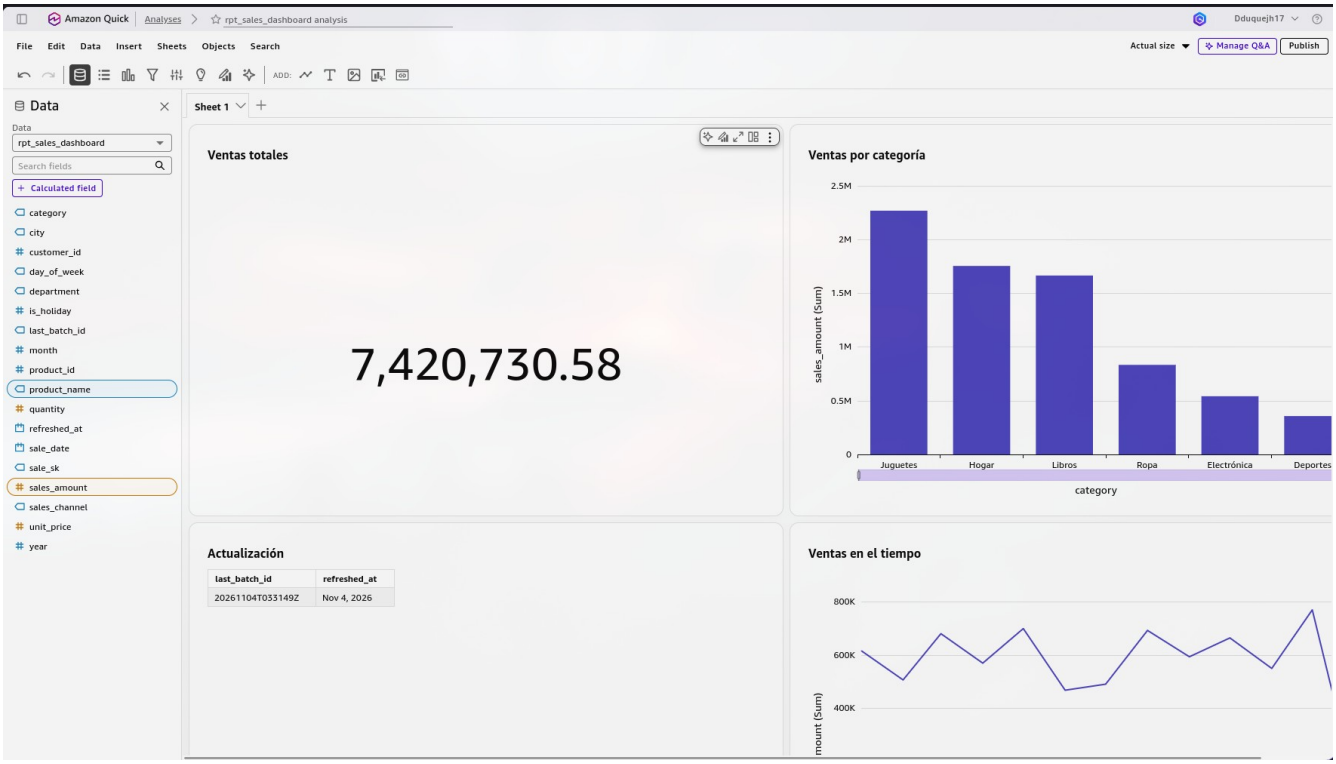
airflow_password_command = "sudo grep AIRFLOW_ADMIN_PASSWORD /opt/ecommerce-dw/.airflo
w-secrets"
airflow_url = "http://52.14.230.30:8080"
bucket_name = "ecommerce-dw-573992724610"
docdb_endpoint = "ecommerce-dw-docdb.cluster-c1aymgwc63hv.us-east-2.docdb.amazonaws.co
m"
instance_id = "i-0faffc839c910f9b8"
mysql_endpoint = "ecommerce-dw-mysql.c1aymgwc63hv.us-east-2.rds.amazonaws.com"
pipeline_config_parameter = "/ecommerce-dw/pipeline/env"
public_ip = "52.14.230.30"
quicksight_vpc_connection_arn = "arn:aws:quicksight:us-east-2:573992724610:vpcConnecti
on/ecommerce-dw-redshift"
redshift_copy_role_arn = "arn:aws:iam::573992724610:role/ecommerce-dw-redshift-s3"
redshift_endpoint = "ecommerce-dw.573992724610.us-east-2.redshift-serverless.amazonaws
.com"
secret_arns = {
  "docdb" = "arn:aws:secretsmanager:us-east-2:573992724610:secret:rds!cluster-3dda92fb
-051b-4e40-b4ae-eba7dcdfe0d1-DZdkdT"
  "mysql" = "arn:aws:secretsmanager:us-east-2:573992724610:secret:rds!db-9b0180c8-b0e6
-40e1-a1d6-9c8dc61957a6-laiRnW"
  "redshift" = "arn:aws:secretsmanager:us-east-2:573992724610:secret:redshift!ecommerc
e-dw-dw_admin-8rtk00"
}
```

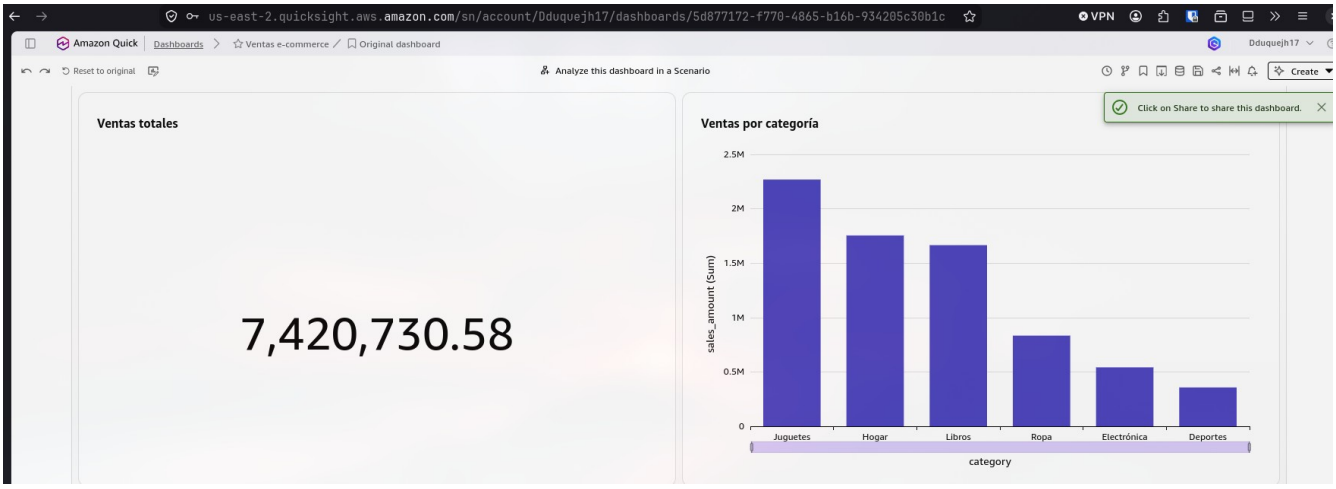
```
◄ 1m57s ◎ terraform output -raw redshift_endpoint
ecommerce-dw.573992724610.us-east-2.redshift-serverless.amazonaws.com⏎

◄ 0s ◎ terraform output secret_value_commands
{
  "docdb" = "aws secretsmanager get-secret-value --secret-id 'arn:aws:secretsmanager:u
s-east-2:573992724610:secret:rds!cluster-3dda92fb-051b-4e40-b4ae-eba7dcdfe0d1-DZdkdT'
--region us-east-2 --profile admin --query SecretString --output text"
  "mysql" = "aws secretsmanager get-secret-value --secret-id 'arn:aws:secretsmanager:u
s-east-2:573992724610:secret:rds!db-9b0180c8-b0e6-40e1-a1d6-9c8dc61957a6-laiRnW' --reg
ion us-east-2 --profile admin --query SecretString --output text"
  "redshift" = "aws secretsmanager get-secret-value --secret-id 'arn:aws:secretsmanage
r:us-east-2:573992724610:secret:redshift!ecommerce-dw-dw_admin-8rtk00' --region us-eas
t-2 --profile admin --query SecretString --output text"
}

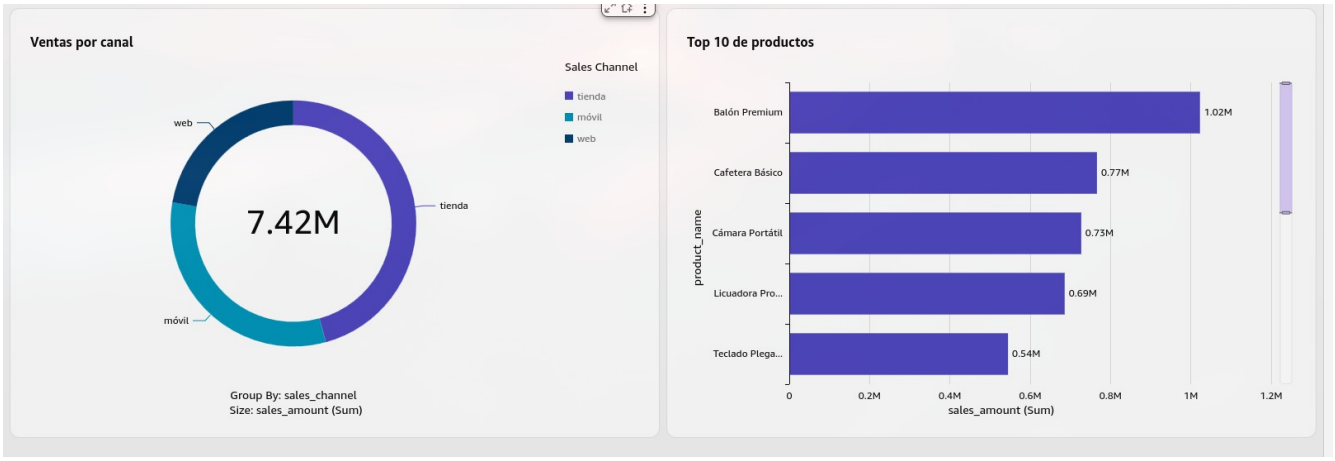
◄ 0s ◎ |    (base) □ Taller-DataWarehouse-DE-20262/terraform △|○◦|via ⎚  default  22:52
```


Amazon Quick
Analyses
rpt_sales_dashboard analysis
Dduquejh17
File
Edit
Data
Insert
Sheets
Objects
Search
Actual size
Manage Q&A
Publish
Data
Data
rpt_sales_dashboard
Search fields
+ Calculated field
category
city
customer_id
day_of_week
department
is_holiday
last_batch_id
month
product_id
product_name
quantity
refreshed_at
sale_date
sale_sk
sales_amount
sales_channel
unit_price
year
Visuals
+ Add
Build
CHANGE VISUAL TYPE
ADD DATA
Add a dimension or measure
Sheet 1
AutoGraph
Add 1 or more fields to build a visual.

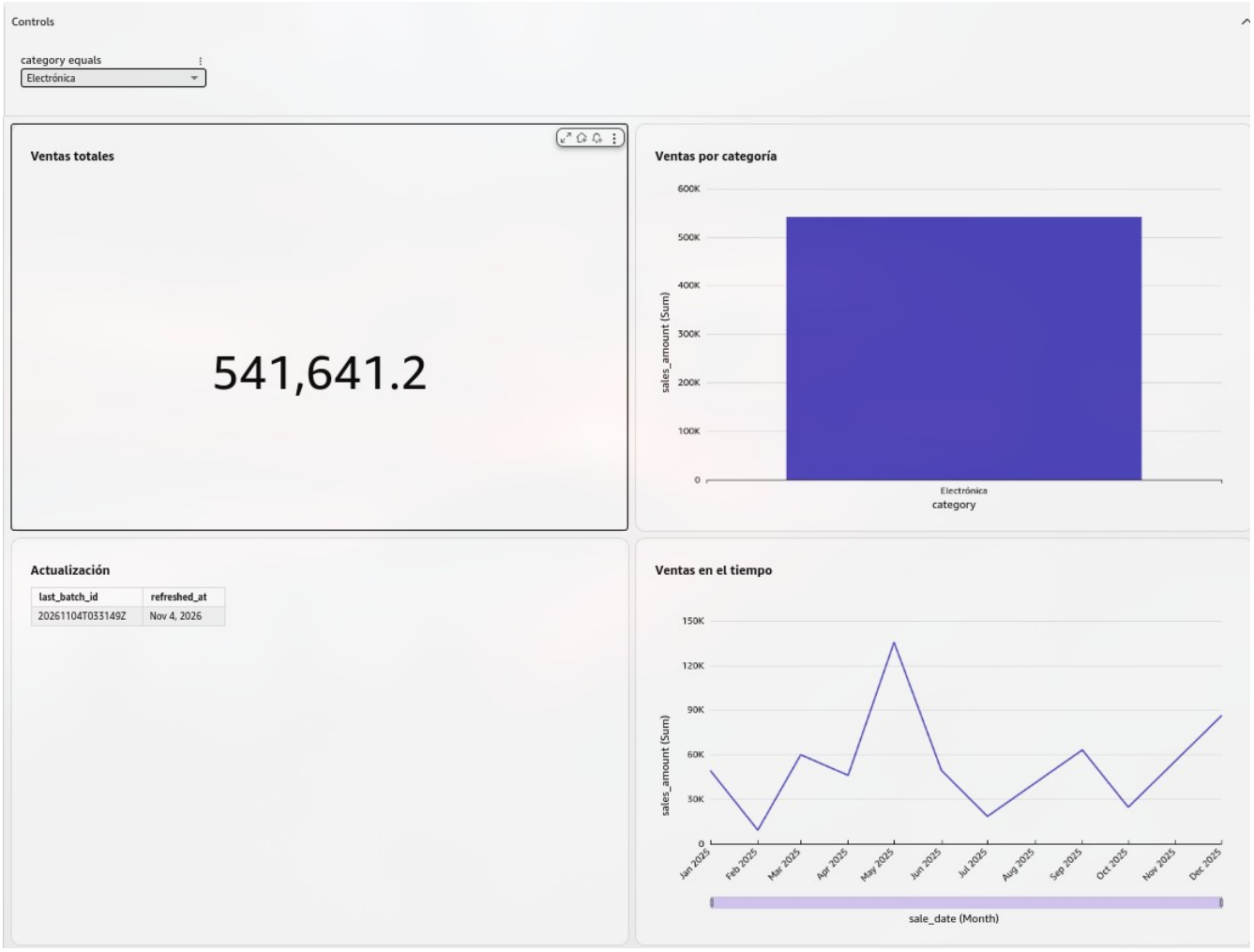

Amazon Quick
Analyses
rpt_sales_dashboard analysis
Dduquejh17
File
Edit
Data
Insert
Sheets
Objects
Search
Actual size
Manage Q&A
Publish
Data
Data
rpt_sales_dashboard
Search fields
+ Calculated field
category
city
customer_id
day_of_week
department
is_holiday
last_batch_id
month
product_id
product_name
quantity
refreshed_at
sale_date
sale_sk
sales_amount
sales_channel
unit_price
year
Sheet 1
Ventas totales
7,420,730.58
Ventas por categoría
2.5M
2M
1.5M
1M
0.5M
0
sales_amount (Sum)
Juguetes
Hogar
Libros
Ropa
Electrónica
Deportes
category
Actualización
last_batch_id
refreshed_at
20261104T033149Z
Nov 4, 2026
Ventas en el tiempo
800K
600K
400K
mount (Sum)





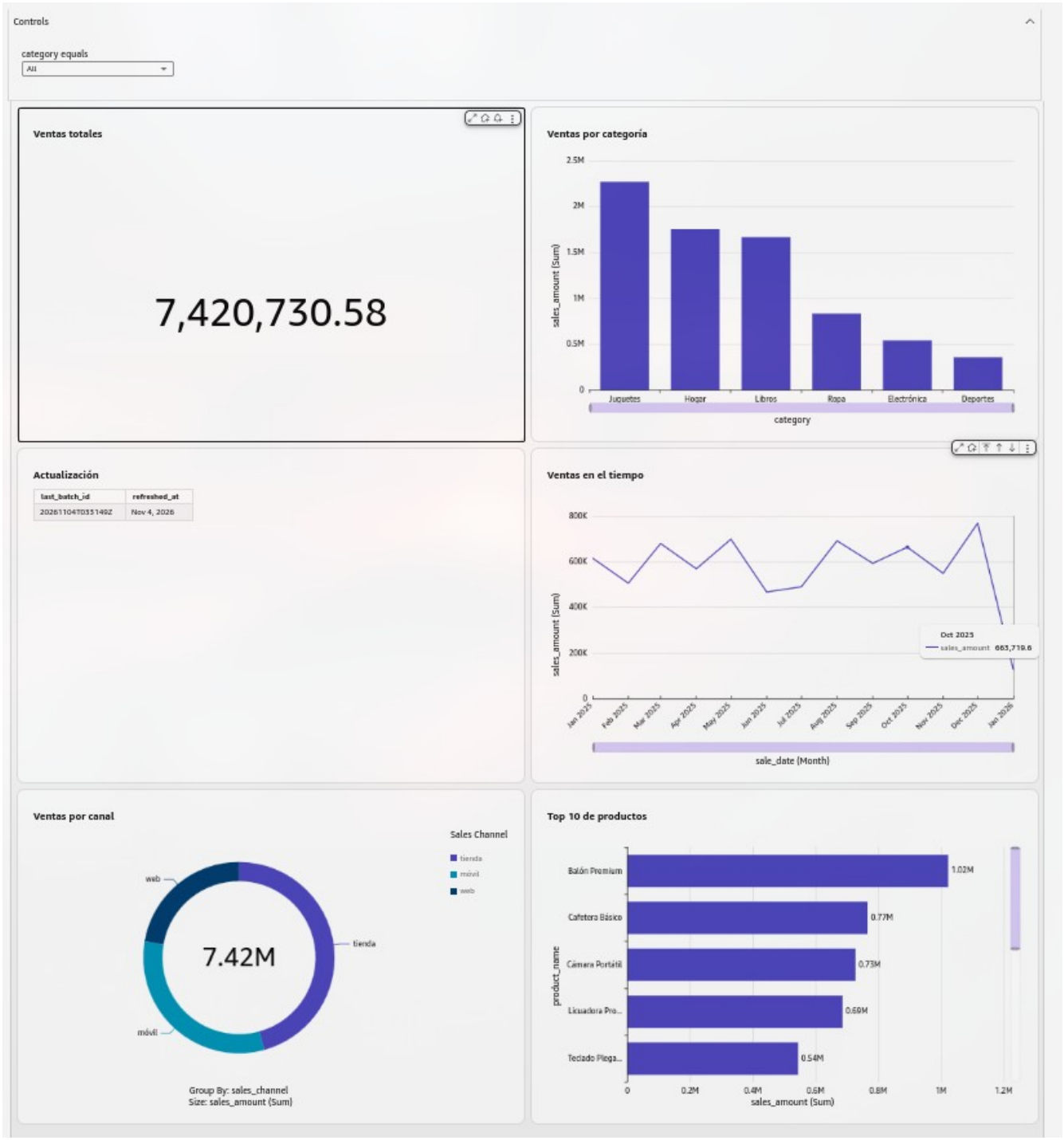


Con filtro aplicado por categoría (electrónica)

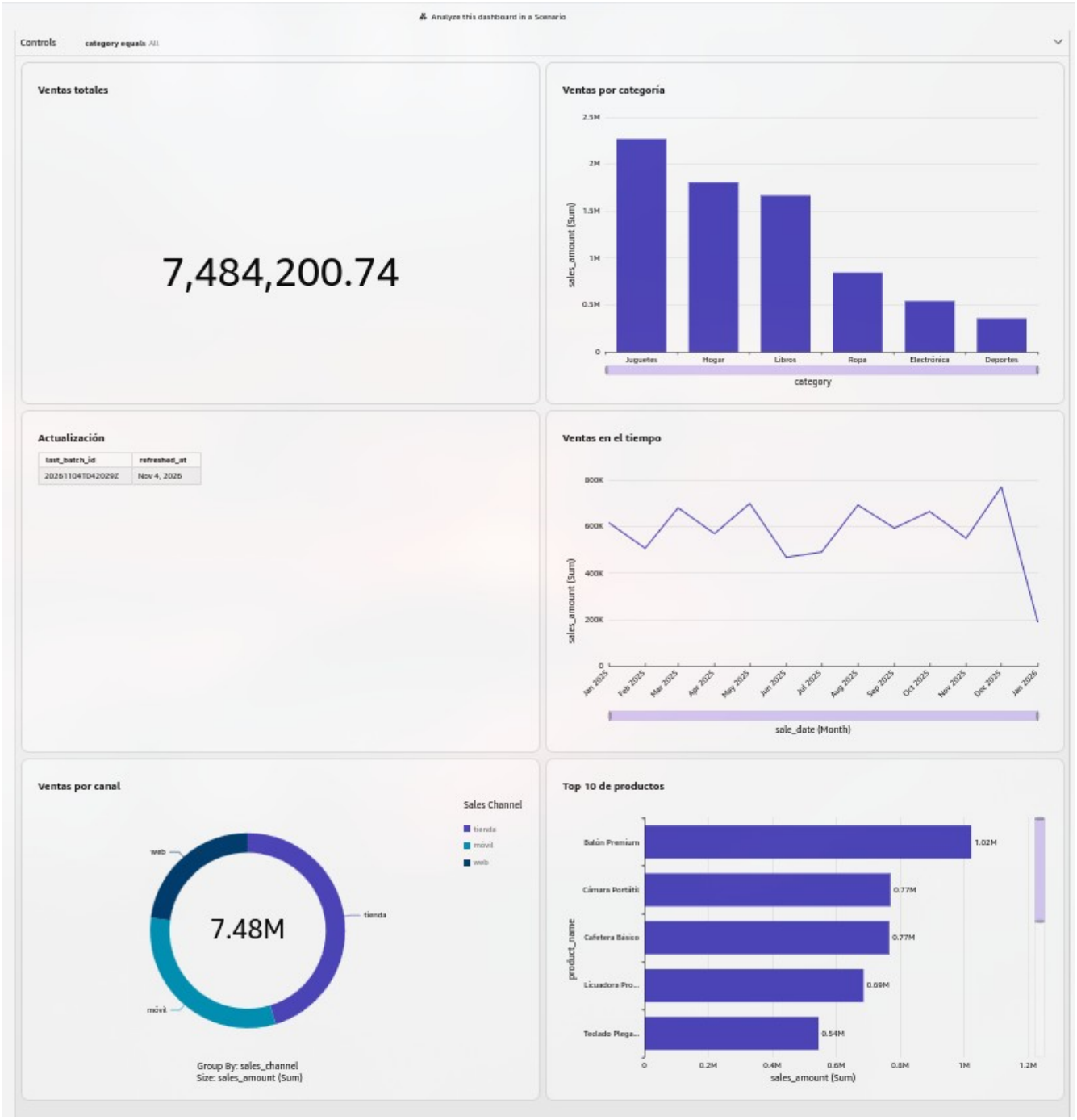



```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/warehouse.sh query "select last_batch_id, refreshed_at, sum(sales_amount) as ventas_totales, count(*) as lineas from marts.rpt_sales_dashboard whe
re category = 'Electrónica' group by last_batch_id, refreshed_at"
last_batch_id     refreshed_at         ventas_totales  lineas
---------------   -------------------  --------------  ------
20261104T033149Z  2026-11-04 03:45:57  541641.20       22
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

Demostración: nuevo lote hasta el dashboard



```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/append_batch.sh --batch-id 4
Agregando a las fuentes el lote diario 4...
/opt/airflow/src/ecommerce_dw/connections.py:99: UserWarning: You appear to be connected to a DocumentDB cluster. For more information regarding feature compatibility and support please visit ht
tps://www.mongodb.com/supportability/documentdb
  return MongoClient(**options)
Batch 4 (2026-01-04): 3 new orders, 3 reviews and 3 navigation events upserted
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/run_pipeline.sh
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
DAG ecommerce_pipeline disparado: run_id=manual__20261104T042021 conf={"mode": "auto", "reprocess_batch_id": ""}
Resultado de manual__20261104T042021: success
batch_id     dag_run_id                                  mode      reprocess  status   failed_tasks  raw_rows  dbt_tests_passed  dbt_tests_failed  finished_at
-----------  ------------------------------------------  --------  ---------  -------  ------------  --------  ----------------  ----------------  -------------------
20261104T042029Z  manual__20261104T042021              auto      False      success  None          12        107               None              2026-11-04 04:22:24
20261104T033149Z  manual__20261104T033141              auto      False      success  None          0         107               None              2026-11-04 03:45:58
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```



Seguridad

- **Secretos:** el repositorio no contiene secretos. Secrets Manager genera y guarda las contraseñas de las bases, y las de Airflow se generan en la instancia.
- **Red:** RDS, DocumentDB y Redshift están en subredes privadas, sin acceso público ni ruta a internet.
- **Cifrado:** las conexiones a las bases exigen TLS, y RDS, DocumentDB y S3 cifran los datos almacenados.
- **Permisos:** la instancia usa su rol IAM por IMDSv2, sin llaves de acceso, y Redshift lee S3 con un rol de solo lectura sobre raw/.
- **S3:** el bucket tiene bloqueo de acceso público, versionado y una política que exige TLS, y el pipeline no puede borrar en raw/.

Limitación conocida: la interfaz de Airflow se sirve por HTTP sin cifrar en el puerto 8080, por lo que solo se abre a una IP autorizada.

Costos

Estimación para us-east-2, con precios on-demand aproximados:

Recurso	Precio aproximado	8 h de trabajo (USD)	Un mes 24/7 (USD)
EC2 t3.medium, IPv4 y 30 GiB	0,047/h + 2,4/mes	0,40	36
RDS db.t4g.micro y 20 GiB	0,016/h + 2,3/mes	0,15	14
DocumentDB db.t4g.medium	0,08/h	0,65	60
Redshift Serverless, 8 RPU	0,375/RPU-hora, solo en uso	1 a 3	Según el uso
Secrets Manager (3 secretos)	0,40/secreto-mes	0,01	1,2
S3, endpoint gateway y SSM	Céntimos	~0	<1
QuickSight (Author)	24/mes, con 30 días de prueba	—	24

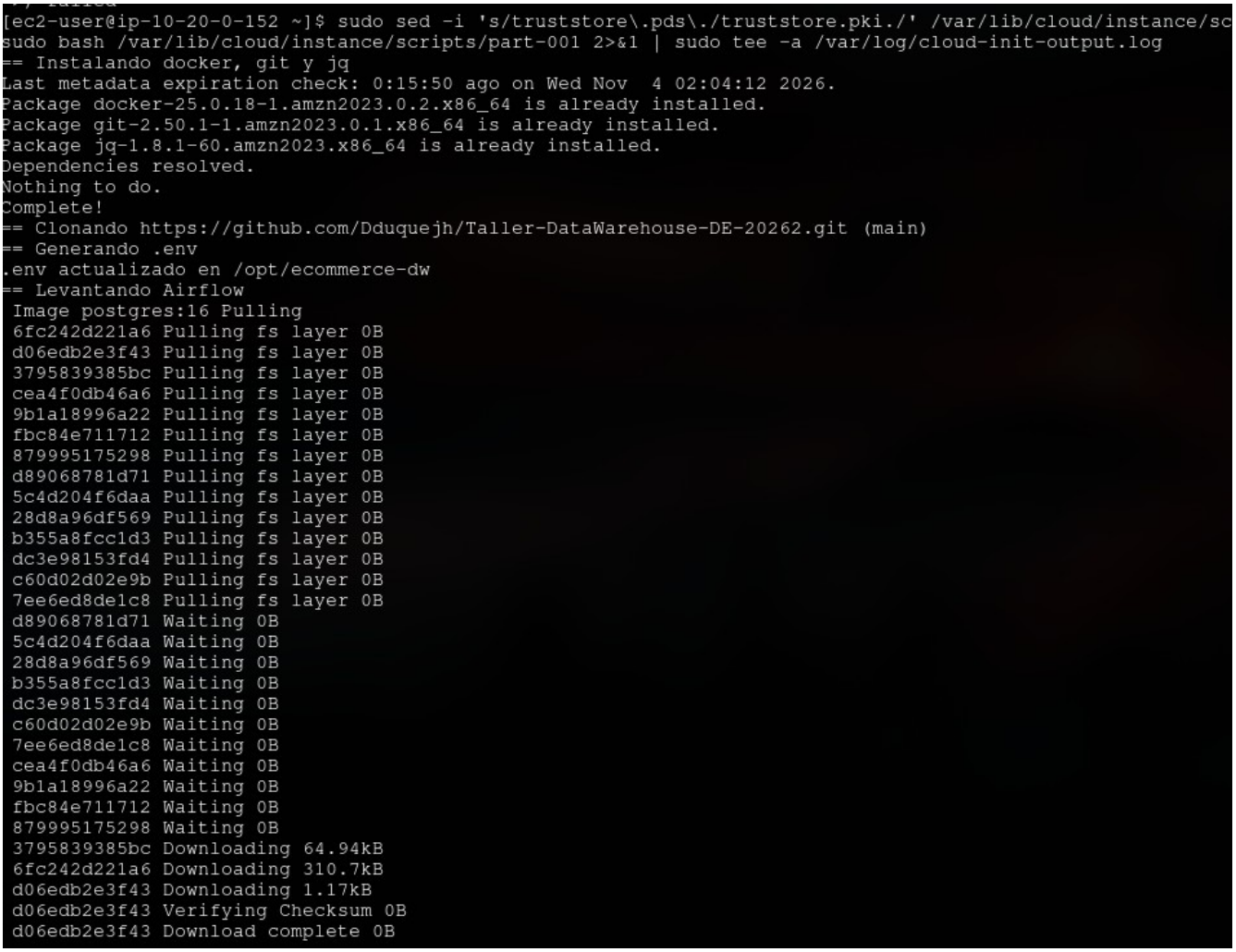
Una sesión de 8 horas cuesta entre USD 2 y 4, sin QuickSight. Redshift es la parte variable: el límite de 16 RPU-hora al día, unos USD 6, desactiva el workgroup si se alcanza, lo que acota el costo a unos USD 7 por sesión. Para controlar el gasto, los recursos se eliminan con terraform destroy al terminar.

Problemas encontrados y correcciones

Al terminar el despliegue, la infraestructura quedó creada, pero la interfaz de Airflow no respondía después de varios minutos. Al revisar el log de arranque de la instancia, se encontró que el script se detuvo al descargar el certificado TLS de AWS, porque el dominio de la URL estaba mal escrito: truststore.pds.rds.amazonaws.com en lugar de truststore.pki.rds.amazonaws.com. Como el dominio no existe, la descarga falló y el script terminó antes de levantar Airflow, como se ve en el siguiente log:

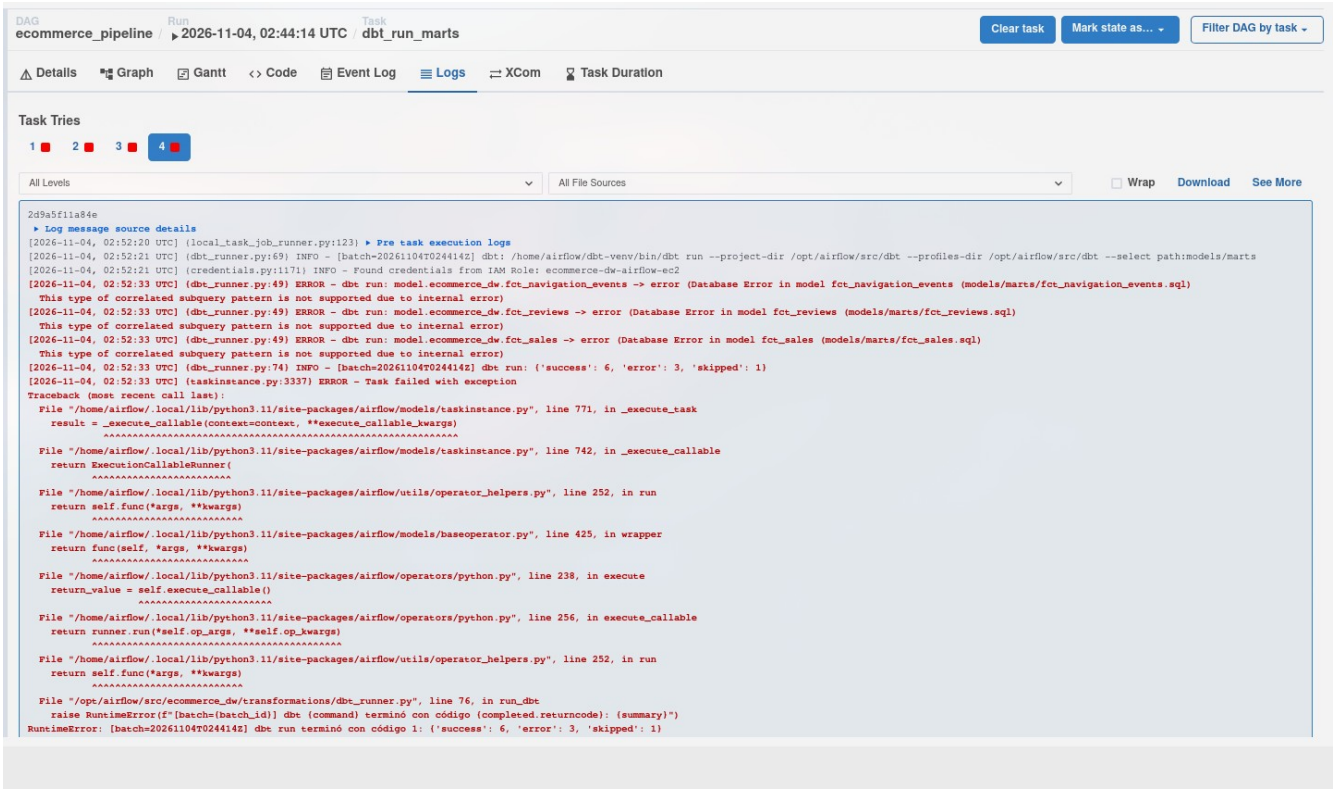


Para solucionarlo, se corrigió la URL en la plantilla terraform/user_data.sh.tftpl, de modo que los próximos despliegues no presenten este error. En la instancia ya creada, se corrigió el dominio en el script de arranque y se volvió a ejecutar, sin recrear la infraestructura. En la siguiente captura se observa que esta vez el script avanza más allá del punto donde falló y comienza a levantar Airflow, descargando las imágenes de Docker:





La primera ejecución del pipeline terminó correctamente, pero la segunda falló en la tarea dbt_run_marts. Los tres hechos (fct_sales, fct_reviews y fct_navigation_events) devolvieron el error de Redshift This type of correlated subquery pattern is not supported, como se ve en el siguiente log:



La primera ejecución no falló porque dbt construye los hechos desde cero y no aplica el filtro incremental. A partir de la segunda, ese filtro incluye un EXISTS correlacionado para detectar filas cuyas dimensiones cambiaron, y Redshift rechaza ese patrón en esta consulta. Por eso, cualquier ejecución posterior a la primera habría fallado. El error no aparecía en la versión local porque DuckDB sí admite esa consulta.

Para solucionarlo, se reemplazó el EXISTS en la macro incremental_candidate_filter por la comparación de una huella de cada fila, calculada con la llave y las columnas seguidas, que también tiene en cuenta los valores nulos. Antes de aplicar el cambio, se ejecutó en Redshift el SQL generado por dbt: la consulta original falló en los tres hechos y la alternativa funcionó. Además, se comprobó que la huella detecta las filas modificadas y no marca las que no cambiaron.



Al ejecutar el script de verificación del lote incremental, el proceso se detuvo con el mensaje jq: parse error: Invalid numeric literal, justo después de disparar el DAG, como se ve en la siguiente salida:

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/verify_incremental_load.sh --batch-id 1
fact                 rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
-------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales            253   253            7293790.26  5bc34ba9198c602c  253               7293790.26
fct_reviews          150   150            0           448b4091a60aea7e  150               0
fct_navigation_events  500   500            0           16b064acaf7659cf  500               0
OK: cada hecho coincide con su fuente validada y no tiene llaves duplicadas
Estado inicial:
fct_sales 253 253 7293790.26 5bc34ba9198c602c
fct_reviews 150 150 0 448b4091a60aea7e
fct_navigation_events 500 500 0 16b064acaf7659cf

/opt/airflow/src/ecommerce_dw/connections.py:99: UserWarning: You appear to be connected to a DocumentDB cluster. For more information regarding feature compatibility and support please visit ht
tps://www.mongodb.com/supportability/documentdb
  return MongoClient(**options)
Agregando a las fuentes el lote diario 1...
Batch 1 (2026-01-01): 3 new orders, 3 reviews and 3 navigation events upserted
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
DAG ecommerce_pipeline disparado: run_id=manual__20261104T031030 conf={"mode": "auto", "reprocess_batch_id": ""}
jq: parse error: Invalid numeric literal at line 1, column 15
[ec2-user@ip-10-20-0-152 ecommerce-dw]$
```

El error no estaba en el pipeline, sino en la parte de scripts/run_pipeline.sh que espera el resultado de la ejecución. El script consulta el estado con airflow dags list-runs -o json, pero Airflow escribe líneas de log antes del JSON en la misma salida. Como jq no podía interpretar esas líneas, el script se detenía y no llegaban a correr las comprobaciones siguientes, aunque el DAG seguía ejecutándose en Airflow.

Para solucionarlo, se filtraron las líneas de log antes de pasar la salida a jq, y se hizo que un fallo puntual de esa consulta no interrumpa la espera. Como el lote 1 ya se había agregado a las fuentes y el script exige un lote nuevo, la verificación se repitió con el lote 2.

```
[ec2-user@ip-10-20-0-152 ecommerce-dw]$ scripts/verify_incremental_load.sh --batch-id 2
fact                 rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
-------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales            256   256            7357260.42  8896681228489ff3  256               7357260.42
fct_reviews          153   153            0           4fac51cf49505be9  153               0
fct_navigation_events  503   503            0           0cad4c3b99974cb9  503               0
OK: cada hecho coincide con su fuente validada y no tiene llaves duplicadas
Estado inicial:
fct_sales 256 256 7357260.42 8896681228489ff3
fct_reviews 153 153 0 4fac51cf49505be9
fct_navigation_events 503 503 0 0cad4c3b99974cb9

/opt/airflow/src/ecommerce_dw/connections.py:99: UserWarning: You appear to be connected to a DocumentDB cluster. For more information regarding feature compatibility and support please visit ht
tps://www.mongodb.com/supportability/documentdb
  return MongoClient(**options)
Agregando a las fuentes el lote diario 2...
Batch 2 (2026-01-02): 3 new orders, 3 reviews and 3 navigation events upserted
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
  warnings.warn(
/home/airflow/.local/lib/python3.11/site-packages/airflow/metrics/base_stats_logger.py:22 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by
default from Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
/home/airflow/.local/lib/python3.11/site-packages/airflow/triggers/base.py:27 RemovedInAirflow3Warning: Timer and timing metrics publish in seconds were deprecated. It is enabled by default from
 Airflow 3 onwards. Enable timer_unit_consistency to publish all the timer and timing metrics in milliseconds.
DAG ecommerce_pipeline disparado: run_id=manual__20261104T031526 conf={"mode": "auto", "reprocess_batch_id": ""}
Resultado de manual__20261104T031526: success
batch_id          dag_run_id                        mode  reprocess  status   failed_tasks  raw_rows  dbt_tests_passed  dbt_tests_failed  finished_at
----------------  --------------------------------  ----  ---------  -------  ------------  --------  ----------------  ----------------  -------------------
20261104T031534Z  manual__20261104T031526            auto  False      success  None          12        107               None              2026-11-04 03:17:37
20261104T031040Z  manual__20261104T031030            auto  False      success  None          12        107               None              2026-11-04 03:12:39
fact                 rows  distinct_keys  total       digest           valid_source_rows  valid_source_total
-------------------  ----  -------------  ----------  ---------------  -----------------  ------------------
fct_sales            259   259            7420730.58  bd515a5603aa1419  259               7420730.58
fct_reviews          156   156            0           8b57ea5592874612  156               0
fct_navigation_events  506   506            0           ce67e63a11c81b5a  506               0
OK: cada hecho coincide con su fuente validada y no tiene llaves duplicadas
OK lote nuevo: cada hecho creció en 3 filas
fct_sales 259 259 7420730.58 bd515a5603aa1419
fct_reviews 156 156 0 8b57ea5592874612
fct_navigation_events 506 506 0 ce67e63a11c81b5a

/opt/airflow/src/ecommerce_dw/connections.py:99: UserWarning: You appear to be connected to a DocumentDB cluster. For more information regarding feature compatibility and support please visit ht
tps://www.mongodb.com/supportability/documentdb
  return MongoClient(**options)
Agregando a las fuentes el lote diario 2...
Batch 2 (2026-01-02): 0 new orders, 3 reviews and 3 navigation events upserted
/home/airflow/.local/lib/python3.11/site-packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or chardet (6.0.0.post1)/charset_normalizer (3.4.5) doesn't match a support
ed version!
```

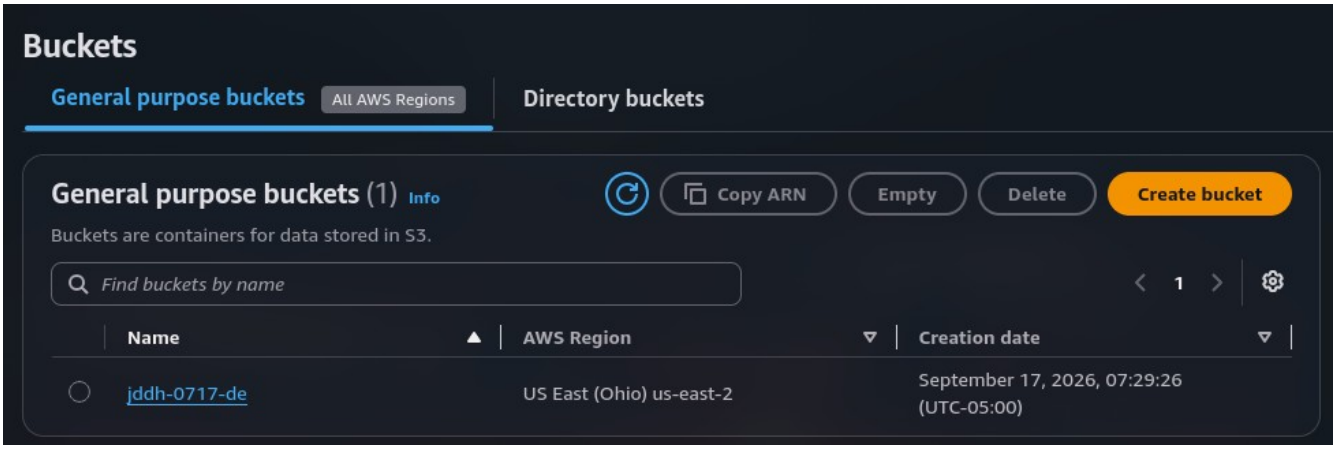
Eliminación de recursos

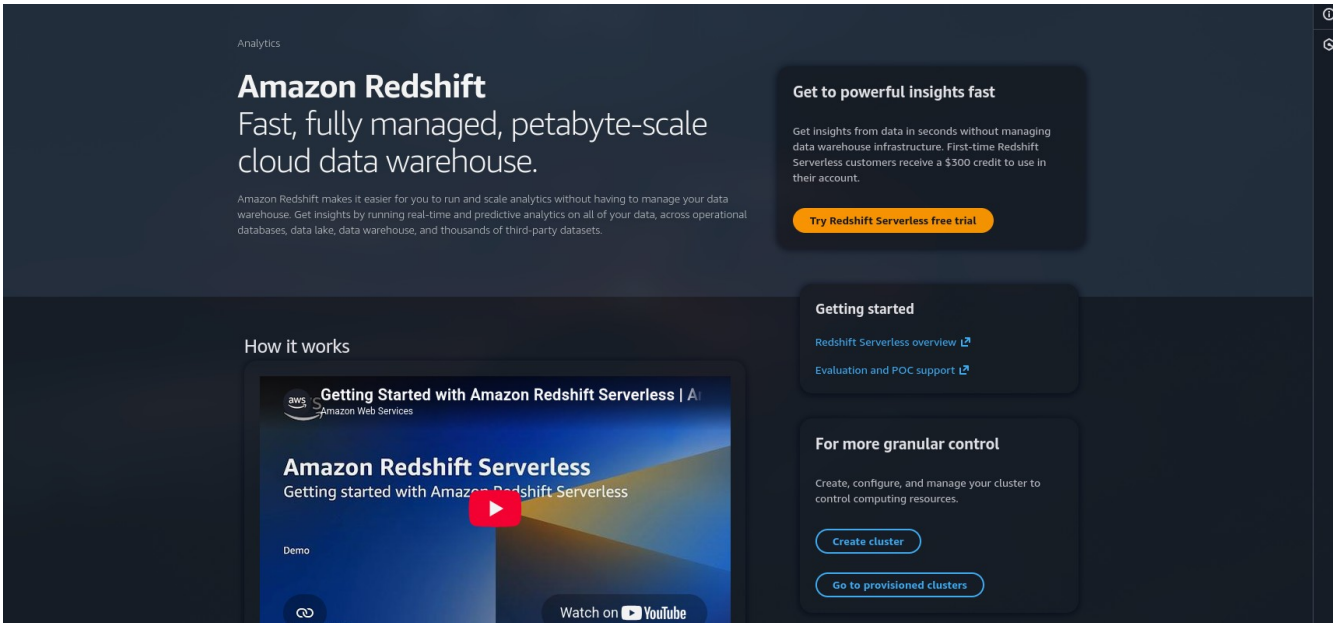
```
aws_docdb_cluster_instance.docdb: Destruction complete after 14m5s
aws_docdb_cluster.docdb: Destroying... [id=ecommerce-dw-docdb]
aws_docdb_cluster.docdb: Still destroying... [id=ecommerce-dw-docdb, 00m10s elapsed]
aws_docdb_cluster.docdb: Still destroying... [id=ecommerce-dw-docdb, 00m20s elapsed]
aws_docdb_cluster.docdb: Still destroying... [id=ecommerce-dw-docdb, 00m30s elapsed]
aws_docdb_cluster.docdb: Still destroying... [id=ecommerce-dw-docdb, 00m40s elapsed]
aws_docdb_cluster.docdb: Still destroying... [id=ecommerce-dw-docdb, 00m50s elapsed]
aws_docdb_cluster.docdb: Destruction complete after 51s
aws_docdb_cluster_parameter_group.docdb: Destroying... [id=ecommerce-dw-docdb50]
aws_docdb_subnet_group.docdb: Destroying... [id=ecommerce-dw-docdb]
aws_security_group.docdb: Destroying... [id=sg-069be91f9546fb3f4]
aws_docdb_subnet_group.docdb: Destruction complete after 0s
aws_subnet.private[1]: Destroying... [id=subnet-0a0a75cc259e422b6]
aws_subnet.private[2]: Destroying... [id=subnet-0050f96e20d7cf02a]
aws_subnet.private[0]: Destroying... [id=subnet-0f6d4376e63bdce87]
aws_docdb_cluster_parameter_group.docdb: Destruction complete after 0s
aws_subnet.private[1]: Destruction complete after 1s
aws_subnet.private[0]: Destruction complete after 1s
aws_subnet.private[2]: Destruction complete after 1s
aws_security_group.docdb: Destruction complete after 1s
aws_vpc.main: Destroying... [id=vpc-06c1941af7c48e45e]
aws_vpc.main: Destruction complete after 1s

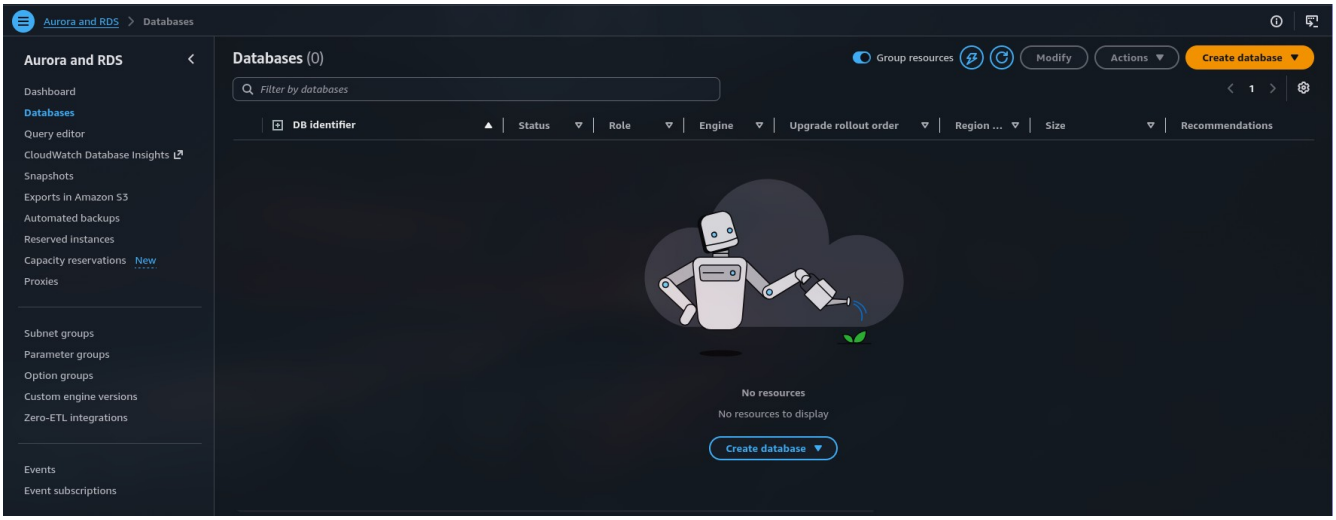
Destroy complete! Resources: 56 destroyed.

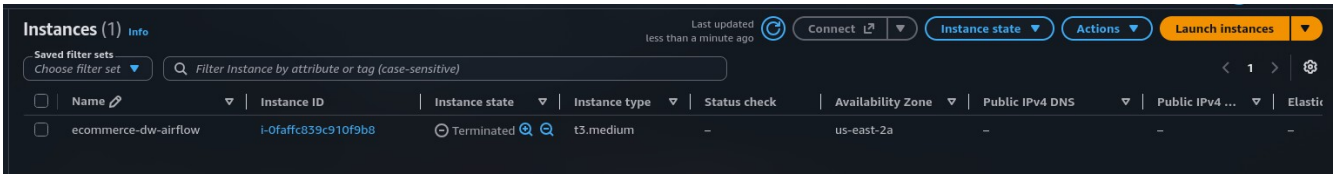
                                                                    ▲228 ▽147

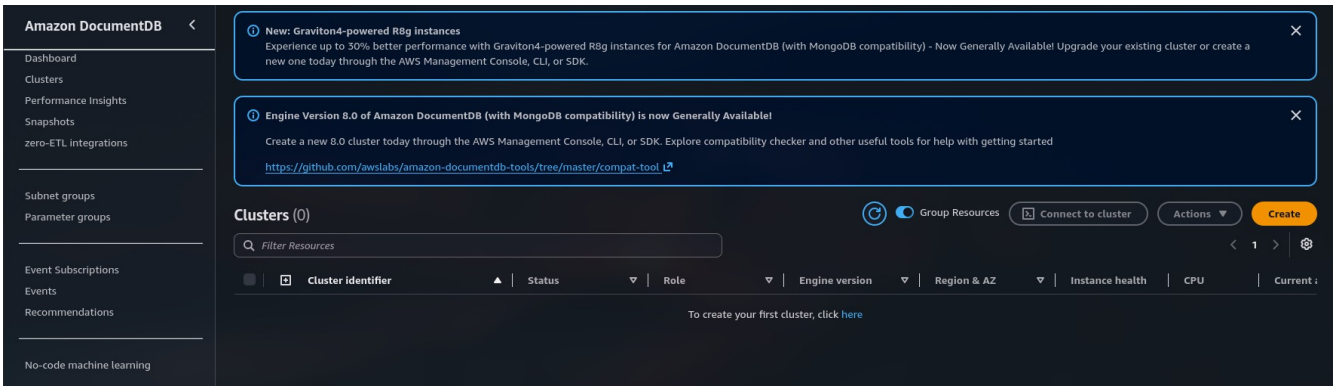
◄ 16m14s ◎
```

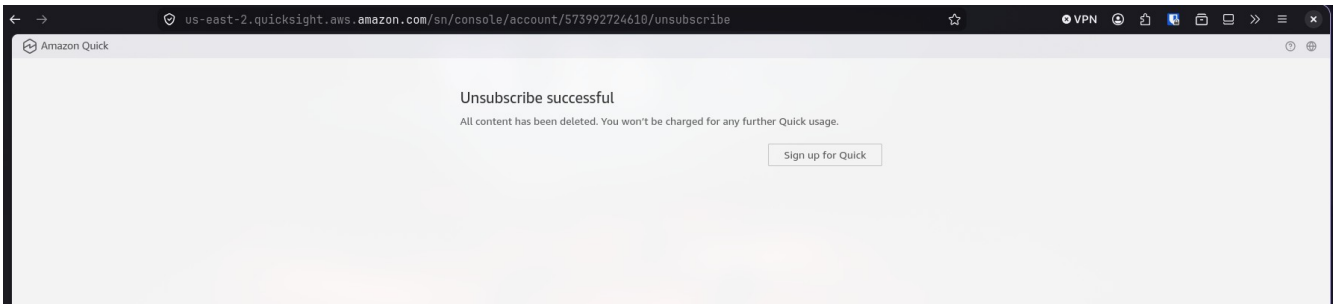

Buckets
General purpose buckets
All AWS Regions
Directory buckets
General purpose buckets (1) Info
Buckets are containers for data stored in S3.
Copy ARN
Empty
Delete
Create bucket
Find buckets by name
1
Name
AWS Region
Creation date
jddh-0717-de
US East (Ohio) us-east-2
September 17, 2026, 07:29:26 (UTC-05:00)


Analytics
Amazon Redshift
Fast, fully managed, petabyte-scale cloud data warehouse.
Amazon Redshift makes it easier for you to run and scale analytics without having to manage your data warehouse. Get insights by running real-time and predictive analytics on all of your data, across operational databases, data lake, data warehouse, and thousands of third-party datasets.
Get to powerful insights fast
Get insights from data in seconds without managing data warehouse infrastructure. First-time Redshift Serverless customers receive a $300 credit to use in their account.
Try Redshift Serverless free trial
How it works
Getting Started with Amazon Redshift Serverless
Amazon Web Services
Amazon Redshift Serverless
Demo
Watch on YouTube
Getting started
Redshift Serverless overview
Evaluation and POC support
For more granular control
Create, configure, and manage your cluster to control computing resources.
Create cluster
Go to provisioned clusters


Aurora and RDS
Databases
Aurora and RDS
Dashboard
Databases
Query editor
CloudWatch Database Insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Capacity reservations New
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
Events
Event subscriptions
Databases (0)
Group resources
Modify
Actions
Create database
Filter by databases
DB identifier
Status
Role
Engine
Upgrade rollout order
Region ...
Size
Recommendations
No resources
No resources to display
Create database


Instances (1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Public IPv4 ...
ecommerce-dw-airflow
i-0faffc839c910f9b8
Terminated
t3.medium
us-east-2a


Amazon DocumentDB
Dashboard
Clusters
Performance Insights
Snapshots
zero-ETL integrations
Subnet groups
Parameter groups
Event Subscriptions
Events
Recommendations
No-code machine learning
New: Graviton4-powered R8g instances
Experience up to 30% better performance with Graviton4-powered R8g instances for Amazon DocumentDB (with MongoDB compatibility) - Now Generally Available! Upgrade your existing cluster or create a new one today through the AWS Management Console, CLI, or SDK.
Engine Version 8.0 of Amazon DocumentDB (with MongoDB compatibility) is now Generally Available!
Create a new 8.0 cluster today through the AWS Management Console, CLI, or SDK. Explore compatibility checker and other useful tools for help with getting started
https://github.com/awslabs/amazon-documentdb-tools/tree/master/compat-tool
Clusters (0)
Group Resources
Connect to cluster
Actions
Create
Filter Resources
Cluster identifier
Status
Role
Engine version
Region & AZ
Instance health
CPU
To create your first cluster, click here


us-east-2.quicksight.aws.amazon.com/sn/console/account/573992724610/unsubscribe
VPN
Amazon Quick
Unsubscribe successful
All content has been deleted. You won't be charged for any further Quick usage.
Sign up for Quick